

1 The actual wave increment and the source's three compatibility targets

The source is the retained 166-page manuscript. Its SHA-256 is given in these two consecutive segments:

0e779481c4da40bd28d1e642e1d8ca57
447d129610df28dfa5a11e9af8ae228f.

The source operations used here are (8.3)–(8.27) and the complete four-step cycle of Proposition 9.6, on printed pages 89–111. Its coefficient classes are Definitions 6.4–6.5 and Proposition 6.6, printed pages 69–72. The source estimates quoted below are attributed to those statements. The following calculation of the actual candidate targets and all pressure-induced changes is a direct finite calculation. It does not infer source iteration exponents from finite smoothness. The source correction cycle is its viscosity-one cycle. The physical increment identities below separately retain any fixed $\nu_{\text{NS}} > 0$; using another physical viscosity in those identities does not identify the resulting candidate with the source's witness.

1.1 The actual covariance increment before pressure reconstruction

Use right-handed cylindrical coordinates (r, θ, z) . An overbar means the normalized angular average followed by normalized auxiliary Haar average, taken at fixed slow variables before phase evaluation. The old source state has actual zero-angular-mean wave w_* , including all its curl corrections. The new candidate is the proved finite field

$$w_y = \sum_{j,a} y_a(t_{c,j}) Z_{ja}, \quad \pi_y = \sum_{j,a} y_a(t_{c,j}) \Pi_{ja}, \quad \langle w_y \rangle_\theta = \langle \pi_y \rangle_\theta = 0.$$

The Z_{ja}, Π_{ja} are the complete physical shapes from the third-growth quantitative calculation; y_a is the actual coupled pair, with $t_{c,j} = t_b + (r_a/L_{s,j})v_j$. No amplitude factor is removed from an auxiliary average: it can depend on $v_j(Y)$. Write the exact double-averaged covariance increment as

$$C_{ab} = \overline{w_{*,a} w_{y,b} + w_{y,a} w_{*,b} + w_{y,a} w_{y,b}}, \quad a, b \in \{r, \theta, z\}. \quad (1)$$

Equivalently, substitute the displayed finite sum for both occurrences of w_y inside this average. In particular C retains every old-wave cross term and every same-label quadratic term. It is symmetric. All its radial integrals below are finite because the new fields are smooth and supported in a compact interior annulus. The old field need only be restricted to this support.

The old source finite correction state has the two exact mean-velocity constraints

$$\int_0^\infty r^2 \bar{v} dr = 0, \quad \int_0^\infty r \bar{\gamma} dr = 0.$$

These are source (9.10) in physical variables. The present initial addition w_y leaves (β, v, γ) unchanged because its angular mean is zero. The temporal mean operation has zero auxiliary mean, and the first two rows of the five-row operation below give zero increments of these integrals. Thus all specified operations preserve their initially zero values. Preservation, by itself, does not change a nonzero entering value to zero.

The already calculated raw tangential residual moments are

$$M_2 = \partial_z J_2, \quad M_1 = \partial_z J_1, \quad J_2 = \int_0^\infty r^2 C_{z\theta} dr, \quad J_1 = \int_0^\infty r C_{zz} dr.$$

Here the candidate has been added with its zero-angular-mean pressure π_y , so the mean pressure is still the old mean pressure. The source algorithm next reconstructs the mean pressure from the updated velocity. This operation changes the axial residual moment.

To compute it, first let $\mathcal{C}_{ab}$ denote the angular average of the tensor inside (1), retaining its auxiliary dependence. Let $\mathcal{D}_r$ be the physical radial derivative on the extended domain, including the auxiliary phase-map derivative. Its auxiliary average is ordinary ∂_r . Since the mean velocity is unchanged by w_y , subtracting the radial conservative equations gives

$$\Delta g_r = -(\mathcal{D}_r + r^{-1})\mathcal{C}_{rr} - \partial_z \mathcal{C}_{zr} + r^{-1}\mathcal{C}_{\theta\theta}. \quad (2)$$

Indeed the radial transport of a divergence-free velocity is $(\mathcal{D}_r + r^{-1})u_r^2 + r^{-1}\partial_\theta(u_\theta u_r) + \partial_z(u_z u_r) - u_\theta^2/r$. Angular averaging removes the angular derivative and all products containing exactly one old mean factor and one zero-angular-mean wave. The source defines g_r as the negative of the mean radial residual apart from the pressure gradient, which gives exactly the signs in (2). The time and viscous averages of w_y are zero, so no physical viscosity has been dropped in this subtraction. The formula remains true for any fixed $\nu_{\text{NS}} > 0$ in the physical residual.

Set $g = \overline{\Delta g_r}$. Auxiliary derivative integrals vanish, and hence

$$g = -(\partial_r + r^{-1})C_{rr} - \partial_z C_{zr} + r^{-1}C_{\theta\theta}. \quad (3)$$

The increment of the source's pressure defect is therefore the following explicit new scalar, rather than either raw tangential moment:

$$\Delta P = \int_0^\infty g \, dr = \int_0^\infty \frac{C_{\theta\theta} - C_{rr}}{r} \, dr - \partial_z \int_0^\infty C_{zr} \, dr. \quad (4)$$

The equality integrates the derivative $\partial_r C_{rr}$; both endpoints vanish by compact support. The weighted second moment of this same source is

$$\int_0^\infty r^2 g \, dr = \int_0^\infty r(C_{rr} + C_{\theta\theta}) \, dr - \partial_z \int_0^\infty r^2 C_{zr} \, dr. \quad (5)$$

For its first term, integration by parts gives $-\int r^2 \partial_r C_{rr} = 2 \int r C_{rr}$, after which the explicit $-C_{rr}/r$ contribution subtracts one copy. These two identities retain the radial and axial coupling exactly.

1.2 Compact mean pressure and the corrected axial flux moment

Retain the source concentration scale and its full axial derivative:

$$z = q^D \eta, \quad 1 - t = q(1 - \eta^2), \quad A = \frac{1}{2} + h, \quad D = \frac{1}{2} - h, \quad q_z = \frac{2\eta q^{1-D}}{1 - 2h\eta^2}.$$

Choose the source fixed bump $\hat{\rho}$ in its reserved mean interval I_m , with $\int \hat{\rho}(x) \, dx = 1$, and put

$$\rho_{\text{phys}} = q^{-1/2} \hat{\rho}(r/\sqrt{q}), \quad c_{\rho, \text{phys}} = \frac{1}{2} \int r^2 \rho_{\text{phys}} \, dr = \frac{q}{2} \int x^2 \hat{\rho}(x) \, dx.$$

Thus $(c_{\rho, \text{phys}})_z = (q_z/q) c_{\rho, \text{phys}}$; it is generally not constant in z .

For precision the source's compact physical primitive is as follows. The fixed slow cutoff χ_m is zero on the inner collar and one on the outer collar of the active annulus. For an auxiliary-dependent scalar f define

$$\begin{aligned} If(r, Y) &= \int_0^r f(s, Y + (s^{d_r} - r^{d_r})v_r) \, ds, \\ Jf(r, Y) &= \int_0^\infty f(s, Y + (s^{d_r} - r^{d_r})v_r) \, ds, \quad I_c f = If - \chi_m Jf. \end{aligned}$$

Here $d_r = 2((1+h)\rho_g - h\kappa_s)$ is the exact source exponent, $\rho_g = \log(4 - \sqrt{2})/\log(4 + \sqrt{2})$, $v_r = (1, -(\sqrt{2}-1))$, and $\kappa_s = 10^{-5}$. The derivative $\mathcal{D}_r$ of the shifted integral differentiates the integration endpoint while cancelling the derivative of its $-r^{d_r}v_r$ shift. Consequently $\mathcal{D}_r I_c f = f - (\partial_r \chi_m) J f$. Also $\overline{Jf} = \int \overline{f} dr$, by translation invariance of Haar measure. This proves the mean identities used next directly.

The source reconstruction after the actual wave addition changes the physical mean pressure by

$$\Delta p_m = I_c(\Delta g_r - \rho_{\text{phys}} \Delta P).$$

Its source has zero radial auxiliary-mean integral by (4). Therefore

$$\partial_r \overline{\Delta p_m} = g - \rho_{\text{phys}} \Delta P, \quad \int r \overline{\Delta p_m} dr = -\frac{1}{2} \int r^2 g dr + c_{\rho, \text{phys}} \Delta P. \quad (6)$$

The first identity follows from the displayed primitive formula; its cutoff remainder has zero auxiliary average. The second integrates $\partial_r(r^2 \overline{\Delta p_m})$ between the two zero endpoints. The full radial residual increment after this pressure update is $-\rho_{\text{phys}} \Delta P - (\partial_r \chi_m) J(\Delta g_r - \rho_{\text{phys}} \Delta P)$. The second term is retained. The source proves its terminal flatness under its all-order uniform class bounds; finite smoothness alone is not that proof.

Use different symbols for the source's two flux defects and the raw candidate fluxes J_2, J_1 . Their increments are exactly

$$\begin{aligned} \Delta J_\theta &= J_2, \\ \Delta J_z &= J_1 - \frac{1}{2} \int r^2 g dr \\ &= \int r \left(C_{zz} - \frac{1}{2} C_{rr} - \frac{1}{2} C_{\theta\theta} \right) dr + \frac{1}{2} \partial_z \int r^2 C_{zr} dr. \end{aligned} \quad (7)$$

These are the differences of source (8.15): the mean-velocity terms are unchanged and the full wave covariance changes by C . The axial pressure identity (6) now proves the exact moments after pressure reconstruction:

$$M_{2, \text{new}} = \partial_z \Delta J_\theta = M_2, \quad M_{1, \text{new}} = \partial_z (\Delta J_z + c_{\rho, \text{phys}} \Delta P) = M_1 + \partial_z \left(-\frac{1}{2} \int r^2 g dr + c_{\rho, \text{phys}} \Delta P \right). \quad (8)$$

These are increment identities. The old source moments and defects are added to obtain the full updated state. No hypothesis that the old defects vanish was used: identical old mean-velocity terms cancel in the subtraction. The product involving $c_{\rho, \text{phys}}$ stays inside ∂_z , including its nonzero derivative.

1.3 New quantitative budgets for the three actual inputs

Retain the previously proved quantities $\mathcal{W}_e, \mathcal{W}_{e,z}, \mathcal{V}_e, \mathcal{Z}_e$ for $e = 1, 2$. They satisfy

$$\|w_*\|_{[e]} = \mathcal{W}_e, \quad \|\partial_z w_*\|_{[e]} = \mathcal{W}_{e,z}, \quad \|w_y\|_{[e]} \leq \mathcal{V}_e, \quad \|\partial_z w_y\|_{[e]} \leq \mathcal{Z}_e.$$

In particular $\mathcal{V}_e, \mathcal{Z}_e$ retain separately the actual amplitude bounds $Y_1 = (A_0/\mu) \hat{\lambda}_3^{-7/8}$ and $Y_2 = 2A_0 \hat{\lambda}_3^{1/8} / (\sigma_2 \sqrt{n_i})$ and their physical source-band factors. Define merely for the following displayed estimates

$$B_e = 2\mathcal{W}_e \mathcal{V}_e + \mathcal{V}_e^2, \quad D_e = \mathcal{W}_{e,z} \mathcal{V}_e + \mathcal{W}_e \mathcal{Z}_e + \mathcal{V}_e \mathcal{Z}_e.$$

Choose $r_- > 0$ enclosing the radial support of the new fields and their axial derivatives from below. Cauchy–Schwarz on the original weighted product measures gives

$$|\Delta P| \leq r_-^{-2} B_1 + 2r_-^{-1} D_1, \quad |\Delta J_\theta| \leq B_2, \quad |\Delta J_z| \leq B_1 + D_2. \quad (9)$$

Here is the full estimate. The matrix $\text{diag}(-1, 1, 0)$ has operator norm one, so $|C_{\theta\theta} - C_{rr}|$ is bounded after averaging by $2|w_*||w_y| + |w_y|^2$. Replacing r^{-1} by $r_-^{-2}r$ and applying Cauchy–Schwarz gives the first part $r_-^{-2}B_1$ of the pressure bound. Differentiating C_{zr} produces the six actual cross and self product terms. Their integral with weight r^e is at most $2D_e$. Replace the unweighted integral by r_-^{-1} times the r -weighted integral to obtain $2r_-^{-1}D_1$ in (4). The same product estimate without differentiation bounds $\int r^2 C_{z\theta}$ by B_2 . Finally $\text{diag}(-1/2, -1/2, 1)$ has norm one. Apply it to the first integral in (7) to get B_1 ; the last differentiated integral is bounded by $\frac{1}{2}(2D_2) = D_2$. This proves all three claims without replacing the two amplitude components by a new unspecified constant.

1.4 The exact five-row map for these targets

The source map acts on the *current* physical triple (P, J_θ, J_z) , after adding (4) and (7) to the old source triple and reconstructing pressure. It does not take (M_2, M_1) as its input. The five equations, in physical variables, are

$$\begin{aligned} \int r^2 \Delta v \, dr &= 0, & \int r \gamma_d \, dr &= 0, \\ \int \frac{2V}{r} \Delta v \, dr &= -P, \\ \int r^2 (G \Delta v + V \gamma_d) \, dr &= -J_\theta, \\ \int (2r G \gamma_d - r V \Delta v) \, dr &= -J_z. \end{aligned} \quad (10)$$

In the last row there is exactly one factor r multiplying $V \Delta v$. The pressure-source term contributes $-\frac{1}{2} \int r^2 (2V/r) \Delta v \, dr = - \int r V \Delta v \, dr$, which proves that factor rather than treating the fifth row as an independent constraint with an arbitrary weight.

Retain the source's reserved patch $x = r/\sqrt{q} \in I_m$. Its actual summed base satisfies source (5.44) and (8.24):

$$\begin{aligned} V_{\text{phys}} &= q^{-A} a(\eta) x^{-1-2\lambda}, & G_{\text{phys}} &= 0, \\ a(\eta) &= 2^{1/2+\lambda} c_{\text{patch}} (1 + \eta^2)^{-1}, & \lambda > 0, & \quad |a(\eta)| \geq a_0 > 0. \end{aligned}$$

The source fixed λ here is its reserved-profile exponent, not a frequency of the coupled problem. In a fixed Q chart, the same physical base is

$$V_* = (Q/q)^A a(\eta) (\sqrt{Q/q} R)^{-1-2\lambda}, \quad G_* = 0.$$

The factor $(Q/q)^A$ has been retained.

Choose a nonnegative nonzero bump $\varphi_0 \in C_c^\infty(I_m)$ supported near an interior point and a fixed spacing $d_b > 0$ so small that the three profiles

$$\varphi_j(x) = e^{-j d_b} \varphi_0(e^{-j d_b} x), \quad j = 0, 1, 2,$$

have disjoint supports in the patch. This is possible because the patch contains an open interval about that point: choose three distinct scaled centers there, then decrease the support radius below half the minimum separation. The symbol d_b distinguishes this source bump spacing from the coupled physical diffusivity d . For every real p , set

$$\mu_p = \int x^p \varphi_0(x) dx > 0, \quad \int x^p \varphi_j(x) dx = \mu_p e^{j d_b p}.$$

The equality is the substitution $x = e^{j d_b} \xi$ and retains the prefactor in φ_j .

Define the two explicit moment matrices

$$A_\theta = \begin{pmatrix} \mu_2 & \mu_2 e^{2d_b} & \mu_2 e^{4d_b} \\ 2a\mu_{-2-2\lambda} & 2a\mu_{-2-2\lambda} e^{d_b(-2-2\lambda)} & 2a\mu_{-2-2\lambda} e^{2d_b(-2-2\lambda)} \\ -a\mu_{-2\lambda} & -a\mu_{-2\lambda} e^{-2d_b\lambda} & -a\mu_{-2\lambda} e^{-4d_b\lambda} \end{pmatrix},$$

$$A_z = \begin{pmatrix} \mu_1 & \mu_1 e^{d_b} \\ a\mu_{1-2\lambda} & a\mu_{1-2\lambda} e^{d_b(1-2\lambda)} \end{pmatrix}, \quad a = a(\eta).$$

Both inverses exist at every source parameter value. Indeed the angular matrix is a diagonal row factor times the matrix with rows $(1, t_p, t_p^2)$ for $p = 2, -2 - 2\lambda, -2\lambda$ and $t_p = e^{d_b p}$. Subtracting the first row from the other two and factoring their differences gives

$$\det[1, t_p, t_p^2]_{p=p_1, p_2, p_3} = (t_{p_2} - t_{p_1})(t_{p_3} - t_{p_1})(t_{p_3} - t_{p_2}).$$

All three powers are distinct for $\lambda > 0$. The diagonal row factor has determinant

$$-2a^2 \mu_2 \mu_{-2-2\lambda} \mu_{-2\lambda} \neq 0.$$

Likewise

$$\det A_z = a\mu_1 \mu_{1-2\lambda} (e^{d_b(1-2\lambda)} - e^{d_b}) \neq 0.$$

This also gives an explicit finite inverse norm bound $\|A^{-1}\| \leq \|\text{adj } A\|/|\det A|$ for either matrix. No uniform limit as $\lambda \downarrow 0$ is taken.

Use the exact row normalizations

$$P_q = q^{2A} P, \quad J_{\theta, q} = q^{2A-3/2} J_\theta, \quad J_{z, q} = q^{2A-1} J_z,$$

and define the five coefficients by

$$\mathbf{u} = A_\theta^{-1}(0, -P_q, -J_{z, q})^T, \quad \mathbf{s} = A_z^{-1}(0, -J_{\theta, q})^T. \quad (11)$$

Then the actual physical velocity increments are

$$\Delta v = q^{-A} \sum_{j=0}^2 u_j \varphi_j(r/\sqrt{q}), \quad \gamma_d = q^{-A} \sum_{j=0}^1 s_j \varphi_j(r/\sqrt{q}). \quad (12)$$

The first and second velocity moments have factors $q^{3/2-A}$ and q^{1-A} , respectively. The three inhomogeneous rows have factors q^{-2A} , $q^{3/2-2A}$, and q^{1-2A} , respectively. Substituting the original base into (10) and changing variables $r = \sqrt{q} x$ therefore gives exactly the two linear systems (11). This proves all five equations, uniqueness in the selected five-dimensional span, and smooth dependence on the actual target values. For instance $\partial_\xi A^{-1} = -A^{-1}(\partial_\xi A)A^{-1}$ follows by differentiating

$AA^{-1} = I$; repeated product rules give every coefficient derivative with the actual q, η factors retained.

The desired axial increment has zero weighted integral and is auxiliary independent. Its exact compact azimuthal potential and radial increment are

$$\Psi(r, z, t) = \frac{1}{r} \int_0^r s \gamma_d(s, z, t) ds, \quad \Delta\beta = -\partial_z \Psi, \quad \Delta\gamma = (\partial_r + r^{-1})\Psi = \gamma_d.$$

The zero moment in (10) makes Ψ vanish beyond its outer support; its lower-end definition makes it vanish below the inner support. Differentiation proves $(\partial_r + r^{-1})\Delta\beta + \partial_z \Delta\gamma = 0$. The support includes the intervals between bump supports, which still lie within the reserved patch. To display the axial scaling cost explicitly, put $\Psi_j(x) = x^{-1} \int_0^x \xi \varphi_j(\xi) d\xi$. Then

$$\begin{aligned} \Psi &= q^{1/2-A} \sum_{j=0}^1 s_j \Psi_j(x), \\ \Delta\beta &= -q^{1/2-A} \sum_{j=0}^1 \left[(\partial_z s_j) \Psi_j + \frac{q_z}{q} s_j \left((1/2 - A) \Psi_j - \frac{x}{2} \Psi_j' \right) \right]. \end{aligned} \quad (13)$$

The two individual Ψ_j can have x^{-1} tails. The exact linear relation $\sum_j s_j \mu_1 e^{jdb} = 0$ cancels those tails in their sum and, after differentiation, in (13). No individual compactness was assumed.

1.5 What the five-row operation actually leaves

Let (β, v, γ) be the current mean correction, after whatever preceding wave and mean updates have actually been performed. Keep the base (b, V, G) and the full current wave w fixed during this five-row step. Use the three full current defects as its targets. Write $\mathcal{D}_r$ for the physical extended radial derivative and $\mathcal{D}_t$ for the physical extended time derivative. With $\Delta_0 = \mathcal{D}_r^2 + r^{-1} \mathcal{D}_r + \partial_z^2$, direct subtraction of the full radial source gives

$$\begin{aligned} R_g &:= g_{r,\text{new}} - g_r - \frac{2V}{r} \Delta v \\ &= -\mathcal{D}_t \Delta\beta - (\mathcal{D}_r + r^{-1})(2b\Delta\beta + 2\beta\Delta\beta + (\Delta\beta)^2) \\ &\quad - \partial_z(b\Delta\gamma + G\Delta\beta + \beta\Delta\gamma + \gamma\Delta\beta + \Delta\beta\Delta\gamma) \\ &\quad + \frac{2v\Delta v + (\Delta v)^2}{r} + \nu_{\text{NS}}(\Delta_0 - r^{-2})\Delta\beta. \end{aligned} \quad (14)$$

Here all increments are auxiliary independent, so their $\mathcal{D}_t$ is ordinary ∂_t . The products with the old means still retain their auxiliary dependence. Equation (14) keeps the actual physical viscosity; the source unit-viscosity formula (8.26) is its case $\nu_{\text{NS}} = 1$.

The full updated defects, with pressure recomputed from the new radial source, are exactly

$$\begin{aligned} P_{\text{new}} &= \int \overline{R_g} dr, \\ (J_\theta)_{\text{new}} &= \int r^2 \gamma \Delta v + v \Delta\gamma + \Delta\gamma \Delta v dr, \\ (J_z)_{\text{new}} &= \int r^2 \gamma \Delta\gamma + (\Delta\gamma)^2 dr - \frac{1}{2} \int r^2 \overline{R_g} dr. \end{aligned} \quad (15)$$

To verify these formulas, expand every quadratic product in the definitions of the current and new defects. The radial source increment is $(2V/r)\Delta v + R_g$. The third row of (10) cancels the old P with

the integral of the displayed linear term. The azimuthal flux increment contains the base-linear terms $G\Delta v + V\Delta\gamma$, cancelled by the fourth row because $\Delta\gamma = \gamma_d$. The axial defect contains $2G\Delta\gamma$ and the radial-source contribution $-\frac{1}{2}r(2V\Delta v)$ after integration. These are precisely the fifth row. All remaining products are the terms displayed in (15); the full wave covariance is unchanged during this mean step. This proves the formulas without assuming their right-hand sides vanish.

The two velocity moments remain exactly zero. The new residual moments therefore have the source form

$$\int r^2 \overline{E_{\theta, \text{new}}} dr = \partial_z (J_\theta)_{\text{new}}, \quad \int r \overline{E_{z, \text{new}}} dr = \partial_z ((J_z)_{\text{new}} + c_{\rho, \text{phys}} P_{\text{new}}).$$

The pressure and $c_{\rho, \text{phys}}$ are the actual reconstructed objects. Thus the five equations cancel specified linear defect contributions and expose the remaining nonlinear and differential terms. Equations (14)–(15) supply the next actual inputs rather than a claim of complete cancellation.

1.6 Exact source operation dictionary and the bounds it requires

For the source's estimates, $Q = 2^{-\ell}$, $\varepsilon = Q^h$, $S_* = \ell^2$, $R = r/\sqrt{Q}$, $Z = z/Q^D$, $T = (1-t)/Q$. Use a coarsest applicable common auxiliary torus H as in source Lemma 6.2; different representatives agree by its integer covering maps. The original radial edge weights are

$$\zeta(X) = \exp \left(-\frac{a_a}{\log^2(X/X_a)} - \frac{a_b}{\log^2(X_b/X)} \right),$$

$$\delta(X) = \min\{1, \log(X/X_a), \log(X_b/X)\}, \quad X = \frac{QR^2}{2q},$$

with the source fixed $a_a, a_b > 0$. In the following bounds the band, label, representative and harmonic are held fixed when differentiating. The constants may depend on the finite correction stage and derivative multi-index, but are uniform over bands, labels, points and rectangle copies.

A mean coefficient $f \in M^\alpha$ has no angular dependence, descends to the common torus, extends smoothly by zero outside the active shell, and satisfies for each multi-index I in (R, Z, T, H)

$$|\partial^I f| \leq C_I \varepsilon^\alpha S_*^{b_I} \zeta \delta^{-d_I}.$$

It can have nonzero auxiliary modes and has no rectangle or pulse envelope restriction. A moment coefficient $F(Z, T) \in S^\alpha$ satisfies $|\partial_{Z, T}^I F| \leq C_I \varepsilon^\alpha S_*^{b_I}$. A labelled nonzero harmonic amplitude $a_{\gamma, m} \in W^\alpha$ has no angular dependence, the source slow, transverse and shell support, smooth zero extension across those boundaries, compatible lift representatives, and

$$|\partial^I a_{\gamma, m}| \leq C_I \varepsilon^\alpha S_*^{b_I} \sqrt{\zeta} \delta^{-d_I} \mathcal{P}_\gamma(v), \quad 0 \leq v \leq L_s.$$

The prescribed source phase has $k_\gamma p_\gamma \in \mathbb{Z} \setminus \{0\}$ and $k_\gamma = \lceil \varepsilon^{-1/2} \rceil$. The harmonic set is finite at a fixed stage and independent of band. Equal label–harmonic terms are grouped before testing the bounds. The pulse envelope is the original positive exponential in (7.12), with its proved Gaussian estimates. Actual cut-off waves additionally have support where the source pulse cutoff ψ is supported and smooth zero extension in v . No divisibility by ψ is required.

The source's anisotropic derivative mappings, with the phase factored out, are

$$\partial_{R, Z, T} : C^\alpha \rightarrow C^\alpha, \quad D_r : C^\alpha \rightarrow C^{\alpha - \kappa_s}, \quad D_z = \varepsilon \partial_Z : C^\alpha \rightarrow C^{\alpha+1},$$

$$c_{i_0} N_{i_0} : C^\alpha \rightarrow C^\alpha, \quad -\varepsilon \partial_T : C^\alpha \rightarrow C^{\alpha+1}, \quad C = M, W.$$

Products obey $M^\alpha M^\beta \subset M^{\alpha+\beta}$ and $M^\alpha W^\beta \subset W^{\alpha+\beta}$. Same-label wave products have this summed order, in M for zero output harmonic and W otherwise; distinct supported labels have zero products. The zero-harmonic global M statement here is for the actual temporally cut-off waves. Before that cutoff, the source gives the mean bounds on the labelled rectangle, not a global M assertion. These are source Proposition 6.6, not deductions from the candidate's finite norm bounds.

Every radial target must first carry the correct measure and weight. For $f_* = Q^{a_f} f_{\text{phys}}$,

$$\int R^e \overline{f_*} dR = Q^{a_f - (e+1)/2} \int r^e \overline{f_{\text{phys}}} dr.$$

For the present inputs this reads

$$\begin{aligned} P_* &= Q^{2A} P, & (J_\theta)_* &= Q^{2A-3/2} J_\theta, & (J_z)_* &= Q^{2A-1} J_z, & (c_\rho)_* &= Q^{-1} c_{\rho, \text{phys}}, \\ (M_2)_* &= Q^{2A-1} M_2, & (M_1)_* &= Q^{2A-1/2} M_1. \end{aligned}$$

In particular (8) becomes exactly $\varepsilon \partial_Z \Delta J_{\theta,*}$ and $\varepsilon \partial_Z (\Delta J_{z,*} + c_{\rho,*} \Delta P_*)$. The factors match because $h + D = 1/2$. This is the extra axial factor used in source (9.12); using the raw J_1 after pressure reconstruction would omit part of its actual input.

For clarity, the following are the four ordered operations of source Proposition 9.6, with the full current velocity and pressure reconstructed after each operation.

1. For each supported nonzero harmonic coefficient $f_{\gamma,m}$ of the full current residual, solve the original transverse pulse equation with zero entrance data and source $-f_{\gamma,m}$. In frame coordinates its coefficient matrix is $H_\gamma - m^2 \varepsilon k_\gamma^2 |n_\gamma|^2 I$. Multiply the solved potential and pressure by ψ and take the full curl. Source Proposition 7.2 maps W^α sources to W^α velocity amplitudes and $W^{\alpha+1/2}$ pressure. Proposition 9.1 assigns the remaining supported linear residual order $\alpha + 1/2 - 3\kappa_s$ and retains its Gaussian cutoff tails additively. The actual candidate's full residual formula determines the input coefficients. Following this wave change, replace w_y by the actual total new wave increment in (1), recompute g_r and reconstruct pressure. Its old-wave cross terms must change with the updated wave.
2. Let F_2, F_1 be the auxiliary-averaged physical tangential residuals of that updated state. Their two moments are the updated versions of (8), including the old defects. For each $e = 1, 2$ retain the fixed normalized physical bump $b_e = q^{-(e+1)/2} \widehat{b}_e(r/\sqrt{q})$ with $\int x^e \widehat{b}_e(x) dx = 1$. Form

$$\mathfrak{M}_e = \int r^e F_e dr, \quad \sigma_e = -r^{-e} \int_0^r s^e (F_e(s) - b_e(s) \mathfrak{M}_e) ds.$$

The zero total weighted integral proves compact support and $(\partial_r + e/r) \sigma_e = -F_e + b_e \mathfrak{M}_e$. Use $\Sigma = Q^{2A}(\sigma_2, \sigma_1)$ as the input to the source's signed amplitude derivative (7.31), with its fixed positive primary amplitudes. Source Proposition 7.6 maps an auxiliary-independent M^α stress to $W^{\alpha-1/2}$. Take its actual curl, retain the old-wave cross and quadratic correction tensor (9.13), and reconstruct pressure. The source does not set $b_e \mathfrak{M}_e$ to zero.

If $\mathfrak{M}_e = \partial_z \mathfrak{J}_e$, its bump term satisfies the exact derivative identity

$$b_e \partial_z \mathfrak{J}_e = \partial_z (b_e \mathfrak{J}_e) - (\partial_z b_e) \mathfrak{J}_e, \quad \partial_z b_e = -\frac{q_z}{2q} ((e+1)b_e + r \partial_r b_e).$$

It follows by differentiating the original q -scaled bump. Thus trading the moment bump for an axial tensor flux leaves the displayed radial-profile derivative term. For the axial equation, $\mathfrak{J}_1 = J_z + c_{\rho, \text{phys}} P$, including both product derivatives in (8).

3. For the zero-auxiliary-mean part of the current angularly invariant residual, first convert the physical residual to the fixed source chart. In this step a star denotes that chart representative:

$$E_* = Q^{2A+1/2} E_{\text{phys}}, \quad E_{a,*}^\circ = E_{a,*} - \langle E_{a,*} \rangle_Y,$$

$$(\Delta\beta_*, \Delta v_*, \Delta\gamma_*) = Q^A (\Delta\beta_{\text{phys}}, \Delta v_{\text{phys}}, \Delta\gamma_{\text{phys}}).$$

The source temporal update in these exact units is

$$\Delta v_* = -c_{i_0}^{-1} N_{i_0}^{-1} E_{\theta,*}^\circ, \quad \gamma_{d,*} = -c_{i_0}^{-1} N_{i_0}^{-1} E_{z,*}^\circ, \quad N_{i_0}^{-1} F = \sum_{k \neq 0} \frac{\widehat{F}(k)}{2\pi i v_t \cdot k} e^{2\pi i k \cdot H}.$$

Here $v_t = (\sqrt{2} - 1, 1)$ and $c_{i_0} = T_g^{i_0} Q^{1+h}$, with $c_{i_0}^{-1} \leq CS_*$. The source Diophantine bound gives $\|N^{-1}F\|_{C^m} \leq C_m \|F\|_{C^{m+4}}$. Let $T_{1,*}$ and $A_{1,*}$ be the source's radial potential and its cutoff remainder in this chart. Realize $\gamma_{d,*}$ by $T_{1,*}\gamma_{d,*}$: $\Delta\beta_* = -\varepsilon \partial_Z T_{1,*}\gamma_{d,*}$ and $\Delta\gamma_* = \gamma_{d,*} - A_{1,*}\gamma_{d,*}$. The actual axial remainder $-A_{1,*}\gamma_{d,*}$ stays in the new velocity. The exact normalized cancellations are

$$c_{i_0} N_{i_0} \Delta v_* = -E_{\theta,*}^\circ,$$

$$c_{i_0} N_{i_0} \Delta\gamma_* = -E_{z,*}^\circ - c_{i_0} N_{i_0} A_{1,*} \gamma_{d,*}.$$

Convert every resulting velocity component back by Q^{-A} before reconstructing the physical pressure or taking physical products and defects. In particular its physical angular update is exactly

$$\Delta v_{\text{phys}} = -Q^{-A} c_{i_0}^{-1} Q^{2A+1/2} N_{i_0}^{-1} E_{\theta,\text{phys}}^\circ = -T_g^{-i_0} N_{i_0}^{-1} E_{\theta,\text{phys}}^\circ.$$

The last equality uses $A+1/2 = 1+h$ and the original $c_{i_0} = T_g^{i_0} Q^{1+h}$. It is also $-N_{\text{abs}}^{-1} E_{\theta,\text{phys}}^\circ$, since $N_{\text{abs}} = T_g^{i_0} N_{i_0}$ on the representative. This proves the conversion without applying a normalized inverse directly to a physical residual.

4. Apply (11) to the three defects of this updated state, use (12) and its actual radial component (13), and reconstruct pressure again. The exact remaining defects are (14)–(15). Source Lemma 8.7 maps S^α targets to M^α tangential increments and $M^{\alpha+1}$ radial increments. Source Lemma 8.8 gives the remaining defect class $S^{\alpha+0.9-2\kappa_s}$ using the actual bounds $v, \gamma \in M^{0.9}$, $\beta \in M^{1.9}$, $\alpha \geq 0.9$, and its reserved-patch base derivative bounds.

The source cycle assumes its full finite correction-state bounds $w \in W^{1/2}$, $w - w_0^{\text{tan}} \in W^{0.68}$, $v, \gamma, p_m \in M^{0.9}$, $\beta \in M^{1.9}$, with supported nonzero residual order $B_j = 1/2 + \sigma_j$, mean and defect order $C_j^* = 1 + \sigma_j$, and $\sigma_j = 1/5 + j/10$. Its ordered estimate proves gains $B_{j+1} - B_j = C_{j+1}^* - C_j^* = 1/10$. These are requirements and conclusions for the source's construction. The explicit candidate budgets and equations above do not by themselves identify those exponents.

There is a precise finite-domain statement available now. Write the full current residual as $R_{\text{old}} + \Delta R$, retaining the old source state and every term of the computed increment. The added candidate uses finitely many labels, with its cutoffs supported strictly inside the source rectangle and radial annulus. On those compact coefficient supports, $\varepsilon > 0$, $\zeta > 0$, $\delta > 0$ and $\mathcal{P}_\gamma(v) > 0$ have positive minima for each fixed label. By local finiteness of the retained source-label family, only finitely many old labels intersect this fixed compact perturbation support. Every differentiated coefficient of ΔR is finite by its explicit smooth shapes, including the retained old-state cross terms there. For any fixed α and derivative I , the maximum of its derivative divided by $\varepsilon^\alpha \sqrt{\zeta} \mathcal{P}_\gamma$ on these finitely many supports is therefore a concrete finite constant. Extend the finite *increment*

family by zero on unused labels. This proves its finite coefficient inequalities without assigning an asymptotic gain to the amplitude. It does not set any old source coefficient to zero: R_{old} retains its actual locally finite family and its attributed source-class bounds. The finite-sum closure in source Proposition 6.6 combines those bounds with the proved increment bounds for each full current input. The same compact-support argument applies to the added mean sources and the reserved bump operators. Thus the operations above are defined on the actual old-plus-increment inputs. Their constants can contain the inverse of a very small Gaussian envelope or source scale; no uniform control for an increasing family of coupled stages or bands follows from this compactness argument. The original component budgets and the full nonlinear remainders remain the quantities to propagate when investigating such a family.

2 Compact completion of the full averaged residual

This section computes a compact symmetric stress and a compact pressure for the complete averaged residual of the actual finite curl candidate. It retains the original radial bumps depending on $q(z, t)$, both axial fluxes, the radial force, and the full cylindrical connection terms. The result is a stress–pressure representation of a computed force; it does not by itself modify a velocity or identify the constructed stress with a covariance of additional waves.

2.1 The actual extended fields and all three averaged components

Use right-handed cylindrical coordinates $x = (r \cos \theta, r \sin \theta, z)$, with oriented orthonormal basis (e_r, e_θ, e_z) and $r > 0$. The retained finite source state is (u_*, p_*) ; the actual added field and its selected pressure are $(v, \pi) = (w_y, \pi_y)$. On a finite collection of source labels the added potential has the specified form

$$\begin{aligned} \mathcal{A}_y &= \sum_j \text{Re} \left[Q_j^{1/2-A} \frac{i n_j \times t_j}{k_j m_j |n_j|^2} e^{i k_j m_j \Phi_j} \right], & v &= \text{curl}_{\mathcal{D}} \mathcal{A}_y, \\ t_j &= \chi_j B_j y_3(t_{c,j}), & y_3 &= \begin{pmatrix} \Theta_3 \\ \Omega_3 \end{pmatrix}, & t_{c,j} &= t_b + \alpha_j v_j, & n_j^T B_j &= 0, & k_j m_j p_j &\in \mathbb{Z} \setminus \{0\}. \end{aligned} \quad (16)$$

Here y_3 is the actual signed third-growth trajectory, the cutoffs and source frame are the retained functions, and Q_j is constant on each source chart. The chosen pressure is the real harmonic sum with chart coefficient

$$Q_j^{-2A} \frac{i \{n_j^T K_j t_j - (n'_j)^T t_j\}}{k_j m_j |n_j|^2} e^{i k_j m_j \Phi_j}.$$

All coefficient components are independent of θ ; their angular dependence is the displayed nonzero integer harmonic. Their radial and axial supports lie in a fixed compact set with r bounded away from zero on the finite time interval under consideration. Smooth zero extension at each source cutoff is retained.

The source absolute auxiliary coordinate is $Y \in \mathbb{T}^2$ and the physical evaluation is

$$\begin{aligned} Y(r, t) &= v_r r^{d_r} + v_t t \pmod{\mathbb{Z}^2}, & v_r &= (1, -b_g), & v_t &= (b_g, 1), & b_g &= \sqrt{2} - 1, \\ \rho_g &= \frac{\log(4 - \sqrt{2})}{\log(4 + \sqrt{2})}, & \kappa_s &= 10^{-5}, & d_r &= 2((1 + h)\rho_g - h\kappa_s), \\ \mathcal{D}_r &= \partial_r + d_r r^{d_r-1} v_r \cdot \partial_Y, & \mathcal{D}_t &= \partial_t + v_t \cdot \partial_Y, & \mathcal{D}_z &= \partial_z. \end{aligned} \quad (17)$$

The angular derivative is ∂_θ . The exponent h here is the unchanged source chart exponent. These lifted derivatives commute, obey the product rule, and satisfy $\mathcal{D}_r r = 1$. Equivalently in Cartesian coordinates they are $\partial_{x_i} + \sum_\alpha (\partial_{x_i} Y_\alpha) \partial_{Y_\alpha}$; their commutators vanish because mixed derivatives of Y commute. Thus evaluation at the physical map preserves every differentiated identity below.

The actual finite source has its retained representation $u_* = \text{curl}_{\mathcal{D}} \mathcal{A}_* + B_* e_\theta$, with $\partial_\theta B_* = 0$. Commutation of the lifted derivatives proves $\text{div}_{\mathcal{D}} \text{curl}_{\mathcal{D}} = 0$. Also $\text{div}_{\mathcal{D}}(B_* e_\theta) = 0$. Consequently both u_* and v are divergence free at every independent Y , not only after physical evaluation.

Let the normalized averages act on cylindrical components before evaluation:

$$\langle f \rangle_\theta = \frac{1}{2\pi} \int_0^{2\pi} f d\theta, \quad \langle f \rangle_Y = \int_{\mathbb{T}^2} f dY, \quad \langle f \rangle = \langle \langle f \rangle_\theta \rangle_Y, \quad \int_{\mathbb{T}^2} 1 dY = 1.$$

Every cylindrical curl operation retains the nonzero angular harmonic in (16). For example its radial and axial components are $r^{-1} \partial_\theta \mathcal{A}_z - \partial_z \mathcal{A}_\theta$ and $(\mathcal{D}_r + r^{-1}) \mathcal{A}_\theta - r^{-1} \partial_\theta \mathcal{A}_r$. Integrating the integer harmonic proves

$$\langle v \rangle_\theta = 0, \quad \langle \pi \rangle_\theta = 0 \quad \text{at every } (r, z, t, Y). \quad (18)$$

This statement includes the complete coefficient curl and every cutoff derivative, rather than only the principal amplitude.

For a smooth periodic coefficient f , integration of its auxiliary derivatives gives

$$\langle \mathcal{D}_r f \rangle = \partial_r \langle f \rangle, \quad \langle \mathcal{D}_t f \rangle = \partial_t \langle f \rangle, \quad \langle \partial_z f \rangle = \partial_z \langle f \rangle, \quad \langle \partial_\theta f \rangle = 0. \quad (19)$$

Applying the first identity twice also treats $\mathcal{D}_r^2$; the derivative of its variable coefficient has zero auxiliary average because that coefficient is independent of Y .

At arbitrary fixed physical viscosity $\nu_{\text{NS}} > 0$, the exact residual increment is

$$\Delta R = \mathcal{D}_t v + (u_* \cdot \nabla_{\mathcal{D}}) v + (v \cdot \nabla_{\mathcal{D}}) u_* + (v \cdot \nabla_{\mathcal{D}}) v - \nu_{\text{NS}} \Delta_{\mathcal{D}} v + \nabla_{\mathcal{D}} \pi. \quad (20)$$

This is the residual of the actual modified state minus the source state's residual at the same viscosity. Put

$$\bar{u}_* = \langle u_* \rangle_\theta, \quad w_* = u_* - \bar{u}_*, \quad S = \langle w_* \otimes v + v \otimes w_* + v \otimes v \rangle. \quad (21)$$

It is a smooth real symmetric axisymmetric tensor compactly supported in the annular and axial set at issue. Each summand contains a factor of the actual compact field v .

Theorem 1. *The complete averaged residual is the full Cartesian divergence of the actual tensor S :*

$$\langle \Delta R \rangle = F_r e_r + F_2 e_\theta + F_1 e_z = \text{div } S, \quad (22)$$

where

$$\begin{aligned} F_r &= (\partial_r + r^{-1}) S_{rr} - r^{-1} S_{\theta\theta} + \partial_z S_{rz}, \\ F_2 &= (\partial_r + 2r^{-1}) S_{r\theta} + \partial_z S_{z\theta}, \\ F_1 &= (\partial_r + r^{-1}) S_{rz} + \partial_z S_{zz}. \end{aligned} \quad (23)$$

Proof. For divergence-free fields, direct Cartesian component expansion rewrites the three advection terms of (20) as the lifted divergence of $u_* \otimes v + v \otimes u_* + v \otimes v$. The mean-flow cross tensor has zero angular average by (18), even when $\bar{u}_*$ depends on Y . Thus its averaged divergence is zero by (19) and the cylindrical divergence formula.

To check the linear radial component explicitly, let

$$\mathcal{L}_0 = \mathcal{D}_r^2 + r^{-1} \mathcal{D}_r + r^{-2} \partial_\theta^2 + \partial_z^2.$$

The complete cylindrical vector Laplacian is

$$\begin{aligned} (\Delta_{\mathcal{D}} v)_r &= (\mathcal{L}_0 - r^{-2}) v_r - 2r^{-2} \partial_\theta v_\theta, \\ (\Delta_{\mathcal{D}} v)_\theta &= (\mathcal{L}_0 - r^{-2}) v_\theta + 2r^{-2} \partial_\theta v_r, \quad (\Delta_{\mathcal{D}} v)_z = \mathcal{L}_0 v_z. \end{aligned}$$

Each average is zero by (18) and (19). The radial pressure average is $\langle \mathcal{D}_r \pi \rangle = \partial_r \langle \pi \rangle = 0$; the angular and axial components vanish as well. The averaged time derivative is zero. These verifications retain the radial connection term and physical viscosity before taking the average.

Finally, for a symmetric tensor independent of θ in cylindrical components, differentiation of $e_r = (\cos \theta, \sin \theta, 0)$ and $e_\theta = (-\sin \theta, \cos \theta, 0)$ gives exactly the three expressions in (23). One direct verification is to write its Cartesian matrix as $O_\theta S O_\theta^T$, where $O_\theta = [e_r, e_\theta, e_z]$, and use $\partial_x = \cos \theta \partial_r - r^{-1} \sin \theta \partial_\theta$ and $\partial_y = \sin \theta \partial_r + r^{-1} \cos \theta \partial_\theta$. The derivatives of O_θ give $-S_{\theta\theta}/r$ in the radial row, $2S_{r\theta}/r$ in the angular row, and S_{rz}/r in the axial row. Averaging the lifted version replaces $\mathcal{D}_r$ by ∂_r . This proves the full Cartesian identity. $\square$

2.2 The compact fluxes and the original varying radial bumps

For $e = 1, 2$, write $\tau_1 = z$, $\tau_2 = \theta$, and set

$$M_e(z, t) = \int_0^\infty r^e F_e(r, z, t) dr, \quad J_e(z, t) = \int_0^\infty r^e S_{z\tau_e}(r, z, t) dr. \quad (24)$$

Multiplication of the corresponding row of (23) by r^e makes its radial term $\partial_r(r^e S_{r\tau_e})$. The radial boundary values vanish by compact annular support. Therefore

$$M_e = \partial_z J_e, \quad \int_{\mathbb{R}} M_e(z, t) dz = 0 \quad (e = 1, 2). \quad (25)$$

Both J_e are smooth and compactly supported in z . No claim that either M_e vanishes pointwise is used.

Retain the original q -dependent bumps, for $q(z, t) > 0$ on the compact source patch:

$$b_e(r, z, t) = q(z, t)^{-(e+1)/2} \widehat{b}_e(r/\sqrt{q(z, t)}), \quad \widehat{b}_e \in C_c^\infty((0, \infty)), \quad \int_0^\infty x^e \widehat{b}_e(x) dx = 1. \quad (26)$$

The fixed profiles may be taken to be the already selected ones. The change of variable $r = \sqrt{q} x$ proves $\int r^e b_e dr = 1$ for every (z, t) . Choose a fixed annulus containing the supports of S and of the bumps on the relevant compact (z, t) set. This is possible because q has positive lower and finite upper bounds there. Outside the compact axial set where a flux or force is nonzero all the formulas below extend by zero.

The exact derivatives and cumulative bump are

$$\begin{aligned} \partial_z b_e &= -\frac{q_z}{2q} ((e+1)b_e + r \partial_r b_e), \\ \beta_e(r, z, t) &= \int_0^r s^e b_e(s, z, t) ds = \int_0^{r/\sqrt{q}} x^e \widehat{b}_e(x) dx, \\ \partial_z \beta_e &= -\frac{q_z}{2q} r^{e+1} b_e(r, z, t), \quad \int_0^\infty r^e \partial_z b_e dr = 0. \end{aligned} \quad (27)$$

The first identity is the product and chain rule. The second is the same radial substitution as the normalization. Differentiating its upper limit proves the third. Differentiating the normalization under the uniformly compact radial integral proves the fourth. In particular none of these identities assigns a sign to q_z .

2.3 Corrected radial primitives and the complete symmetric tensor

Define the actual corrected primitives

$$\sigma_e(r, z, t) = -r^{-e} \int_0^r s^e (F_e(s, z, t) - \partial_z(b_e(s, z, t)J_e(z, t))) ds, \quad e = 1, 2. \quad (28)$$

Their exact radial equations are

$$(\partial_r + e/r)\sigma_e = -F_e + \partial_z(b_e J_e). \quad (29)$$

This follows by multiplying (28) by r^e and using the fundamental theorem of calculus.

For comparison with the initially proposed primitive, put

$$\sigma_e^{\text{old}} = -r^{-e} \int_0^r s^e (F_e - b_e \partial_z J_e) ds.$$

Every missing term can be displayed exactly:

$$\sigma_e = \sigma_e^{\text{old}} + r^{-e} J_e \partial_z \beta_e = \sigma_e^{\text{old}} - \frac{r q_z}{2q} b_e J_e. \quad (30)$$

Indeed expanding $\partial_z(b_e J_e)$ in (28) proves the first equality, and (27) proves the second. With the old primitive the angular or axial stress divergence below would instead be $-F_e - (\partial_z b_e)J_e$. Thus the correction is necessary for the retained varying bump. It vanishes in the special case $q_z = 0$.

There is an exact expression entirely in the original tensor entries:

$$\begin{aligned} C_e(r, z, t) &= \int_0^r s^e S_{z\tau_e}(s, z, t) ds - \beta_e(r, z, t) J_e(z, t), \\ \sigma_e &= -S_{r\tau_e} - r^{-e} \partial_z C_e. \end{aligned} \quad (31)$$

To prove it, integrate the corresponding component of (23) only up to r , obtaining $\int_0^r s^e F_e ds = r^e S_{r\tau_e} + \partial_z \int_0^r s^e S_{z\tau_e} ds$. Substitute this identity and $\int_0^r s^e b_e ds = \beta_e$ into (28). Below both radial supports $C_e = 0$; above both supports its two summands are J_e and J_e , so it is again zero. This proves its compact support as well as the exact relation of the constructed radial stress to the actual covariance tensor.

Theorem 2. *The real symmetric axisymmetric tensor with cylindrical entries*

$$\begin{aligned} T_{rr} &= 0, & T_{r\theta} &= T_{\theta r} = \sigma_2, & T_{rz} &= T_{zr} = \sigma_1, \\ T_{z\theta} &= T_{\theta z} = -b_2 J_2, & T_{zz} &= -b_1 J_1, & T_{\theta\theta} &= rF_r + r\partial_z \sigma_1 \end{aligned} \quad (32)$$

is smooth and compactly supported in an annular cylinder and obeys the full Cartesian identity

$$\text{div } T = -(F_r e_r + F_2 e_\theta + F_1 e_z). \quad (33)$$

If the term rF_r is omitted only in $T_{\theta\theta}$, the exact divergence is instead $-(0, F_2, F_1)$ in cylindrical components.

Proof. The integral of the numerator in (28) over the entire radial half-line is

$$M_e - \partial_z \left(J_e \int_0^\infty s^e b_e ds \right) = \partial_z J_e - \partial_z J_e = 0.$$

Thus the numerator primitive is zero below its support and zero above it. Division by r^e introduces no singularity because it is zero on a fixed neighborhood of the axis. All derivatives under its fixed compact radial integral exist, and the normalization identity holds smoothly in (z, t) , so its extension by zero is smooth. Compact axial support follows because both F_e and J_e vanish outside the fixed compact axial set. The same statements hold for all entries of (32). Passing to the Cartesian matrix $O_\theta T O_\theta^T$ preserves smoothness away from the axis; near the axis it is identically zero. Hence this is a smooth compactly supported Cartesian tensor.

Apply the full symmetric axisymmetric divergence formula already proved in (23). Its three rows are

$$\begin{aligned} (\operatorname{div} T)_r &= -\frac{rF_r + r\partial_z\sigma_1}{r} + \partial_z\sigma_1 = -F_r, \\ (\operatorname{div} T)_\theta &= (\partial_r + 2/r)\sigma_2 - \partial_z(b_2J_2) = -F_2, \\ (\operatorname{div} T)_z &= (\partial_r + 1/r)\sigma_1 - \partial_z(b_1J_1) = -F_1. \end{aligned}$$

The last two identities are (29), including every derivative of b_e . This proves the Cartesian vector identity and, on omitting rF_r , its stated radial-zero variant. $\square$

2.4 Compact pressure, tracefree stress, and the exact gauge relation

The trace and its prescribed pressure are explicitly

$$\operatorname{tr} T = rF_r + r\partial_z\sigma_1 - b_1J_1, \quad p_T = \frac{1}{3}(rF_r + r\partial_z\sigma_1 - b_1J_1), \quad T^0 = T - p_TI_3. \quad (34)$$

Then $\operatorname{tr} T^0 = 0$, and every field is smooth and compactly supported in the same annular cylinder. The exact full cancellation is

$$F + \operatorname{div} T^0 + \nabla p_T = 0. \quad (35)$$

To check its sign, for any scalar p Cartesian differentiation gives $\operatorname{div}(pI_3) = \nabla p$. Therefore $\operatorname{div} T^0 = \operatorname{div} T - \nabla p_T = -F - \nabla p_T$, which proves the displayed equation. In components the new diagonal entries are

$$T_{rr}^0 = -p_T, \quad T_{\theta\theta}^0 = rF_r + r\partial_z\sigma_1 - p_T, \quad T_{zz}^0 = -b_1J_1 - p_T.$$

The axial stress divergence alone changes by $-\partial_z p_T$; the pressure gradient contributes $+\partial_z p_T$. Thus the combined stress–pressure correction preserves the exact tangential equations. There is no claim that their stress term alone stays unchanged.

More generally $T - pI_3$ with any smooth compact scalar p satisfies $F + \operatorname{div}(T - pI_3) + \nabla p = 0$. The choice in (34) is the unique one making this stress tracefree. The map from symmetric tensors to tracefree-tensor-scalar pairs,

$$T \mapsto \left(T - \frac{1}{3} \operatorname{tr} T I_3, \frac{1}{3} \operatorname{tr} T \right),$$

has exact inverse $(U, p) \mapsto U + pI_3$ when $\operatorname{tr} U = 0$. It intertwines div with $(U, p) \mapsto \operatorname{div} U + \nabla p$. This supplies the full map between the two presentations, including compact support and signs.

2.5 All total force and torque components and the divergence image

The Cartesian components of the averaged force are

$$F = (F_r \cos \theta - F_2 \sin \theta, F_r \sin \theta + F_2 \cos \theta, F_1).$$

Integration in θ with the physical volume element $dx = r dr d\theta dz$ gives

$$\begin{aligned} \int_{\mathbb{R}^3} F dx &= 2\pi e_z \int_{\mathbb{R}} M_1(z, t) dz = 0, \\ \int_{\mathbb{R}^3} x \times F dx &= 2\pi e_z \int_{\mathbb{R}} M_2(z, t) dz = 0. \end{aligned} \tag{36}$$

For the torque calculation, direct multiplication in the oriented basis gives

$$x \times F = -zF_2e_r + (zF_r - rF_1)e_\theta + rF_2e_z.$$

The first two terms integrate to zero in θ , and the third gives the second equation of (36). Thus all three force and all three torque components vanish. The last equalities use the already proved compact flux identities (25). The same torque vanishes about every fixed origin x_0 , because subtracting x_0 changes it by $-x_0 \times \int F dx = 0$. On a finite axial slab $a < z < b$, the axial force and torque are respectively $2\pi(J_1(b, t) - J_1(a, t))$ and $2\pi(J_2(b, t) - J_2(a, t))$. The quantities need not vanish on an individual slice.

There is also a direct Cartesian stress verification. Since $F_j = -\partial_k T_{jk}$ and T is compact, integration by parts gives

$$\int F_j dx = 0, \quad \int x_i F_j dx = \int T_{ji} dx = - \int S_{ji} dx. \tag{37}$$

Symmetry makes the antisymmetric part of the first-moment matrix zero, which is precisely all three torque components. Its symmetric part is not asserted to vanish. The second equality with S uses $F = \text{div } S$. Equivalently $\text{div}(T + S) = 0$; multiplying this equation by x_i and integrating proves $\int (T + S)_{ji} dx = 0$. In the tracefree-pressure presentation the same first moment is $\int T_{ji}^0 dx + \delta_{ij} \int p_T dx$. All these identities persist under any finite time differentiation on a common compact support, by differentiation under the integral.

For completeness, this construction proves the exact image of compact symmetric-stress divergence in the axisymmetric class. Let an open axial interval I_z contain the actual axial supports, with $q > 0$ smooth on a neighborhood of its relevant compact subsets at the fixed time. In the following two spaces all supports are compact subsets of $\{r > 0, z \in I_z\}$. Thus the original bump is defined everywhere it is used, and integrating between axial support endpoints stays inside I_z . Let $\mathcal{F}_0$ be the space of smooth real axisymmetric compact forces supported away from $r = 0$ and satisfying exactly

$$\int_{\mathbb{R}} \int_0^\infty r F_1 dr dz = 0, \quad \int_{\mathbb{R}} \int_0^\infty r^2 F_2 dr dz = 0.$$

Let $\mathcal{S}_c$ be the space of smooth real compact symmetric axisymmetric Cartesian tensors supported away from $r = 0$. Define $\mathcal{D} : \mathcal{S}_c \rightarrow \mathcal{F}_0$ by $\mathcal{D}(U) = -\text{div } U$. Integration by parts proves that its values satisfy both defining identities of $\mathcal{F}_0$, and axisymmetry makes the other force and torque components vanish as above.

For any $F \in \mathcal{F}_0$, compute its actual radial moments and set

$$J_e(z) = \int_{-\infty}^z \int_0^\infty r^e F_e(r, s) dr ds.$$

This is a smooth compact axial function: its derivative is M_e , it vanishes below the axial support, and the defining zero total integral makes it vanish above that support. Use the fixed bumps (26), the corrected primitive (28), and all entries of (32). The proof of Theorem 2 uses only these identities and proves a compact tensor $\mathcal{C}_b F$ with

$$\mathcal{D}\mathcal{C}_b F = F \quad (F \in \mathcal{F}_0). \quad (38)$$

Thus the two total moments are necessary and sufficient, with an explicit right inverse. For the actual force in (22), this primitive in z equals the already computed J_e from S , because both have derivative M_e and vanish at the negative axial end. No additional compatibility condition has been imposed on the actual candidate.

For fixed q and bump profiles, all operations defining $\mathcal{C}_b$ are linear. Hence $\mathcal{P}_b = \mathcal{C}_b \mathcal{D}$ is a projection on $\mathcal{S}_c$: (38) gives $\mathcal{P}_b^2 = \mathcal{P}_b$ and $\ker \mathcal{P}_b = \ker \mathcal{D}$. In particular the completed actual tensor is $T = \mathcal{P}_b(-S)$, and $T + S$ is a compact divergence-free symmetric tensor. This gives an exact relation to the actual averaged covariance, not merely equality of its two radial moments.

2.6 Exact source chart, bump, stress, and pressure powers

Fix one source chart and retain all of its constants:

$$r = \sqrt{Q} R, \quad z = Q^D Z, \quad 1 - t = QT, \quad A = \frac{1}{2} + h, \quad D = \frac{1}{2} - h, \quad \varepsilon = Q^h. \quad (39)$$

The dyadic label Q is fixed under differentiation. The averaged quantities are independent of Y and θ , so in this subsection the complete normalized spatial derivatives are ∂_R , $R^{-1}\partial_\theta$ and $\varepsilon\partial_Z$, with the same cylindrical basis derivatives as before. Define the exact chart representatives

$$\begin{aligned} S^* &= Q^{2A} S, & F^* &= Q^{2A+1/2} F, & T^* &= Q^{2A} T, & \Sigma_e &= Q^{2A} \sigma_e, \\ J_e^* &= Q^{2A-(e+1)/2} J_e = \int_0^\infty R^e S_{z\tau_e}^* dR, & M_e^* &= Q^{2A-e/2} M_e = \varepsilon \partial_Z J_e^*, \\ q_* &= q/Q, & b_e^* &= Q^{(e+1)/2} b_e(\sqrt{Q}R, z, t) = q_*^{-(e+1)/2} \widehat{b}_e(R/\sqrt{q_*}). \end{aligned} \quad (40)$$

The two moment powers follow directly from $r^e dr = Q^{(e+1)/2} R^e dR$ and $\partial_z = Q^{-D} \partial_Z$; the remaining factor in M_e^* is $Q^{1/2-D} = \varepsilon$. In particular the scaled bump satisfies $\int R^e b_e^* dR = 1$.

Every term of the corrected primitive has the same exact stress unit:

$$\begin{aligned} \Sigma_e &= -R^{-e} \int_0^R X^e (F_e^* - \varepsilon \partial_Z (b_e^* J_e^*)) dX, \\ (\partial_R + e/R) \Sigma_e &= -F_e^* + \varepsilon \partial_Z (b_e^* J_e^*), \\ \Sigma_e &= \Sigma_e^{\text{old}} - \frac{\varepsilon R \partial_Z q_*}{2q_*} b_e^* J_e^*. \end{aligned} \quad (41)$$

To verify the first line, insert $F_e = Q^{-2A-1/2} F_e^*$ and $b_e J_e = Q^{-2A} b_e^* J_e^*$ in (28), and change the integration variable to $s = \sqrt{Q} X$. The second follows by differentiation. For the third, the physical correction in (30) has factor $\sqrt{Q} q_z/q$; using $q_z = Q^{1-D} \partial_Z q_*$ gives $\sqrt{Q} q_z/q = \varepsilon \partial_Z q_*/q_*$. This verifies the original q derivative, including its sign. Also, with $C_e^* = Q^{2A-(e+1)/2} C_e$,

$$\Sigma_e = -S_{r\tau_e}^* - \varepsilon R^{-e} \partial_Z C_e^*.$$

The completed stress and its pressure become exactly

$$\begin{aligned} T_{rr}^* &= 0, & T_{r\theta}^* &= \Sigma_2, & T_{rz}^* &= \Sigma_1, & T_{z\theta}^* &= -b_2^* J_2^*, & T_{zz}^* &= -b_1^* J_1^*, \\ T_{\theta\theta}^* &= RF_r^* + \varepsilon R \partial_Z \Sigma_1, & p_T^* &= Q^{2A} p_T = \frac{1}{3} (RF_r^* + \varepsilon R \partial_Z \Sigma_1 - b_1^* J_1^*), \\ (T^0)^* &= T^* - p_T^* I_3, & F^* + \operatorname{div}_*(T^0)^* + \nabla_* p_T^* &= 0. \end{aligned} \quad (42)$$

For example the radial row is $-T_{\theta\theta}^*/R + \varepsilon \partial_Z \Sigma_1 = -F_r^*$, and the tangential rows use (41). The pressure identity follows from the same Cartesian isotropic-tensor calculation as before. Physical divergence and pressure gradient both have factor $Q^{-2A-1/2}$ because $\nabla_x = Q^{-1/2} \nabla_*$ on the chart. Thus the radial residual, hoop stress, both surviving axial fluxes, varying-bump correction and compact pressure all have their exact source units. No decay or uniform estimate at a singular endpoint is inferred from these finite compact-support identities.

3 Original-scale derivative costs of the actual compatibility defects

The state and increment in this section are exactly the finite source state and actual third-growth curl constructed in the preceding reader. Write $u_* = \langle u_* \rangle_\theta + w_*$, $v = w_y$, and

$$C = \langle w_* \otimes v + v \otimes w_* + v \otimes v \rangle,$$

where the outer brackets are the normalized angular and auxiliary averages before physical evaluation. Every component below is in the original oriented cylindrical orthonormal basis. The full moment completion and source pressure calculation prove the formulas used here; we also derive the scalar compatibility formulas directly to fix all signs. Throughout this section the unadorned symbols P, J_θ, J_z denote the defect increments $\Delta P, \Delta J_\theta, \Delta J_z$ caused by this actual wave addition. They are not the full old-plus-new source defects. The unchanged old defects must be added when applying the source correction to the whole updated state.

3.1 Actual mixed derivatives of the third-growth curl

Retain the fixed finite set of source labels j , the actual clocks $t_{c,j} = t_b + \alpha_j v_j$, and the two full shape fields W_{ja} . The original data t_b, r_a, S_3, u_i are independent of these spatial labels. In particular

$$v = \sum_{j,a} Q_j^{-A} y_a(t_{c,j}) W_{ja}, \quad A = \frac{1}{2} + h, \quad D = \frac{1}{2} - h, \quad \partial_t t_{c,j} = \alpha_j Q_j^{-1-h}, \quad \partial_z t_{c,j} = 0.$$

Let $Y_{1,n} = T_n$ and $Y_{2,n} = O_n$ be the complete finite derivative arrays already proved for the actual coupled solution. Thus $Y_{1,0} = (A_0/\mu) \hat{\lambda}_3^{-7/8}$, $Y_{2,0} = 2A_0 \hat{\lambda}_3^{1/8}/(\sigma_2 \sqrt{n_i})$, $Y_{1,1} = 2\gamma_i Y_{1,0}$, and $Y_{2,1} = \gamma_i A_0 \hat{\lambda}_3^{1/8}/(\sigma_2 \sqrt{n_i})$. Higher entries are those closed recurrences in the retained moving-parent data; no independent amplitude bound is introduced here.

For a pair $I = (a, b) \in \mathbb{N}^2$, put $\partial^I = \partial_z^a \partial_t^b$ and $I_z = I + (1, 0)$. On a fixed chart the full time operator is $\mathbf{t}_{*,j} = -\varepsilon_j \partial_{T_j} + c_{i,j} N_{i,j}$. The exact mixed derivative formula is

$$\begin{aligned} \partial_z^a \partial_t^b v &= \sum_{j,c} \sum_{s=0}^b \binom{b}{s} Q_j^{-A-aD-b(1+h)} \alpha_j^s y_c^{(s)}(t_{c,j}) \\ &\quad \cdot \partial_{Z_j}^a \mathbf{t}_{*,j}^{b-s} W_{jc}. \end{aligned} \quad (43)$$

All operators on the shapes include their phase, frame and cutoff derivatives. The shape time operator is the full derivative after auxiliary evaluation, not a derivative that holds the fast phase

fixed. To prove the identity, differentiate the physical product b times in time. If s of the derivatives act on y_c , its factor is $(\alpha_j Q_j^{-1-h})^s y_c^{(s)}$; the other $b-s$ derivatives give $Q_j^{-(b-s)(1+h)} \mathfrak{t}_{*,j}^{b-s} W_{jc}$. There are $\binom{b}{s}$ such choices. The a axial derivatives act only on the shape and give $Q_j^{-aD} \partial_{Z_j}^a$. The axial derivative commutes with the stated time operator, since its coefficients are fixed label constants. Combining the powers proves (43) without dropping any derivative.

For each $e \in \{-1, 0, 1, 2\}$, define the precise weighted norm

$$\|f\|_{[e]}^2 = \int_0^\infty r^e \langle |f|^2 \rangle dr, \quad \|F\|_{[e],j}^2 = \int_0^\infty R_j^e \langle |F|^2 \rangle dR_j.$$

Every field being integrated is supported in a compact annulus away from zero, so the negative weight introduces no singular integral. Define the explicit actual-shape budget

$$\begin{aligned} \mathcal{V}_{e,(a,b)} = \sum_{j,c} \sum_{s=0}^b \binom{b}{s} \alpha_j^s Y_{c,s} Q_j^{(e+1)/4-A-aD-b(1+h)} \\ \cdot \|\partial_{Z_j}^a \mathfrak{t}_{*,j}^{b-s} W_{jc}\|_{[e],j}. \end{aligned} \quad (44)$$

Then $\|\partial^I v\|_{[e]} \leq \mathcal{V}_{e,I}$. Indeed the measure identity $r^e dr = Q_j^{(e+1)/2} R_j^e dR_j$, valid also for $e = -1$, and the triangle inequality applied to (43) give this bound. There is no inter-label orthogonality assumption. Put $\mathcal{W}_{e,I} = \|\partial^I w_*\|_{[e]}$, restricted to a fixed radial set enclosing the supports of the actual new fields and all derivatives at issue. The old wave includes its existing curl corrections. These are norms of the specified source state, not missing amplitude variables.

3.2 A contraction estimate that retains every product derivative

For $J = (j_1, j_2) \leq I = (a, b)$, set $\binom{I}{J} = \binom{a}{j_1} \binom{b}{j_2}$, and define

$$\begin{aligned} \mathcal{B}_{e,I} = \sum_{J \leq I} \binom{I}{J} (\mathcal{W}_{e,J} \mathcal{V}_{e,I-J} + \mathcal{V}_{e,J} \mathcal{W}_{e,I-J} \\ + \mathcal{V}_{e,J} \mathcal{V}_{e,I-J}). \end{aligned} \quad (45)$$

If E is any fixed real symmetric three-by-three matrix in cylindrical components, then

$$\left| \partial^I \int r^e (E : C) dr \right| \leq \|E\|_{\text{op}} \mathcal{B}_{e,I}. \quad (46)$$

For a single outer product, $E : (f \otimes g) = f^T E g$, whose absolute value is at most $\|E\|_{\text{op}} |f| |g|$. The repeated product rule gives exactly the sum over $J \leq I$ in (45). Cauchy–Schwarz on the original weighted angular-auxiliary-radial measure gives each displayed product of norms. Compact smooth support justifies differentiation under the integral. Axial and time derivatives do not differentiate the cylindrical basis. This proves (46) for every specified order. For the time derivative, the full lifted physical operator is $\mathcal{D}_t = \partial_t + N_{\text{abs}}$, with N_{abs} a constant torus direction. The Haar integral of $N_{\text{abs}} f$ is zero by periodicity. Therefore $\partial_t \langle f \rangle = \langle \mathcal{D}_t f \rangle$. The same identity applies repeatedly, and axial differentiation commutes with this average and this constant torus direction. Consequently the ordinary mixed derivatives of the averaged tensor in (46) are exactly the averages of the full physical derivatives estimated in (43).

3.3 The actual three defects and the pressure-corrected axial flux

Since the wave increment has zero angular mean and is divergence-free, the change of the radial mean pressure source is

$$g = -(\partial_r + r^{-1})C_{rr} - \partial_z C_{zr} + r^{-1}C_{\theta\theta}.$$

The three actual source compatibility-defect increments are therefore

$$\begin{aligned} P &= \int \frac{C_{\theta\theta} - C_{rr}}{r} dr - \partial_z \int C_{zr} dr, \\ J_\theta &= \int r^2 C_{z\theta} dr, \\ J_z &= \int r \left(C_{zz} - \frac{1}{2}C_{rr} - \frac{1}{2}C_{\theta\theta} \right) dr + \frac{1}{2}\partial_z \int r^2 C_{zr} dr. \end{aligned} \tag{47}$$

The first formula is $\int g dr$, using the zero boundary integral of $\partial_r C_{rr}$. For the last, integration by parts gives

$$\int r^2 g dr = \int r(C_{rr} + C_{\theta\theta}) dr - \partial_z \int r^2 C_{zr} dr.$$

Substitute it into the source definition $J_z = \int r C_{zz} dr - \frac{1}{2} \int r^2 g dr$. This proves all three identities in (47). In particular the axial source defect is not replaced by the raw flux $\int r C_{zz} dr$.

Let the source pressure bump be the original

$$\rho(r, z, t) = q^{-1/2} \hat{\rho}(r/\sqrt{q}), \quad \int \hat{\rho}(x) dx = 1, \quad c_0 = \frac{1}{2} \int x^2 \hat{\rho}(x) dx.$$

It has $\int \rho dr = 1$ and $c_\rho = \frac{1}{2} \int r^2 \rho dr = c_0 q$. The auxiliary average of the compact source pressure increment is

$$\overline{\Delta p_m}(r) = \int_0^r (g(s) - \rho(s)P) ds.$$

Its derivative is $g - \rho P$, and its integrand has zero total radial integral, giving compact support. Its weighted pressure moment is $\int r \overline{\Delta p_m} dr = -\frac{1}{2} \int r^2 g dr + c_\rho P$. Here is the exact map to the full auxiliary-dependent source operation. Let Δg_r be the angularly averaged source increment before auxiliary averaging, so that $\langle \Delta g_r \rangle_Y = g$, and put $f = \Delta g_r - \rho P$. The source's shifted primitive uses

$$\begin{aligned} \mathcal{I}f(r, Y) &= \int_0^r f(s, Y + (s^{d_r} - r^{d_r})v_r) ds, \\ \mathcal{J}f(r, Y) &= \int_0^\infty f(s, Y + (s^{d_r} - r^{d_r})v_r) ds, \quad \mathcal{I}_c f = \mathcal{I}f - \chi_m \mathcal{J}f. \end{aligned}$$

The fixed source exponent and direction are the original d_r, v_r , and χ_m is its inner-zero, outer-one radial cutoff. The full pressure increment is $\Delta p_m = \mathcal{I}_c f$. Haar translation invariance gives $\langle \mathcal{J}f \rangle_Y = \int (g - \rho P) dr = 0$ and $\langle \mathcal{I}f \rangle_Y = \int_0^r (g - \rho P) ds$. This proves the asserted averaged pressure formula from the full map. The lifted radial derivative cancels the shift derivative and yields

$$\mathcal{D}_r \Delta p_m - \Delta g_r = -\rho P - (\partial_r \chi_m) \mathcal{J}f.$$

The second term has zero auxiliary average but need not vanish before physical evaluation. No estimate for its singular-time flatness is inferred from its zero average. Consequently the double-averaged axial residual-moment increment after this reconstruction is

$$\partial_z K_z, \quad K_z = J_z + c_0 q P.$$

The full product, including the derivatives of q , is retained.

The following simultaneous quantitative estimates hold at every finite mixed order:

$$\begin{aligned}
|\partial^I P| &\leq \mathcal{B}_{-1,I} + \frac{1}{2}\mathcal{B}_{0,I_z}, \\
|\partial^I J_\theta| &\leq \frac{1}{2}\mathcal{B}_{2,I}, \\
|\partial^I J_z| &\leq \mathcal{B}_{1,I} + \frac{1}{4}\mathcal{B}_{2,I_z}, \\
|\partial^I K_z| &\leq \mathcal{B}_{1,I} + \frac{1}{4}\mathcal{B}_{2,I_z} \\
&\quad + |c_0| \sum_{J \leq I} \binom{I}{J} |\partial^J q| (\mathcal{B}_{-1,I-J} + \frac{1}{2}\mathcal{B}_{0,(I-J)_z}).
\end{aligned} \tag{48}$$

For the first line the diagonal contraction matrix is $\text{diag}(-1, 1, 0)$, of norm one, whereas the rz entry is the contraction against the symmetric matrix with rz, zr entries $1/2$, of norm $1/2$. For the second line use the analogous $z\theta$ matrix. For the third the diagonal matrix is $\text{diag}(-1/2, -1/2, 1)$, of norm one, and the coefficient $1/2$ multiplying C_{zr} contributes an additional factor $1/2$. These facts and (46) prove the first three bounds. The exact product rule for qP proves the fourth. Replacing I by I_z gives bounds for the two residual moments that actually enter the source radial inverse. Thus every pressure, radial and axial coupling has a bound in the original shape fields and third-growth constants.

3.4 Exact derivatives of the original moving scale and bumps

The source scale is determined by

$$q - z^2 q^{2h} = 1 - t, \quad z = q^D \eta, \quad D = \frac{1}{2} - h, \quad 0 < h < 1/100, \quad -1 < \eta < 1.$$

Set $B(\eta) = 1 - 2h\eta^2 > 1 - 2h > 0$. Implicit differentiation of the first equation, followed by the displayed formula for z , gives

$$\begin{aligned}
q_z &= \frac{2\eta q^A}{B}, & q_t &= -\frac{1}{B}, \\
\eta_z &= \frac{q^{-D}(1 - \eta^2)}{B}, & \eta_t &= \frac{D\eta q^{-1}}{B}.
\end{aligned} \tag{49}$$

For example $q_z = 2zq^{2h}/B$, and $D + 2h = A$. Differentiating $\eta = zq^{-D}$ gives the last two formulas; the identity $2D + 2h = 1$ gives the factor $1 - \eta^2$. This proves the derivative dictionary without assigning a sign to q_z .

Put $x = r/\sqrt{q}$. For a smooth function $f(x, \eta)$ define the explicit differential expressions

$$\mathcal{Z}_\sigma f = \frac{2\sigma\eta f - \eta x f_x + (1 - \eta^2)f_\eta}{B}, \quad \mathcal{T}_\sigma f = \frac{-\sigma f + \frac{1}{2}x f_x + D\eta f_\eta}{B}.$$

Equations (49) and the ordinary chain rule prove

$$\partial_z(q^\sigma f) = q^{\sigma-D} \mathcal{Z}_\sigma f, \quad \partial_t(q^\sigma f) = q^{\sigma-1} \mathcal{T}_\sigma f, \quad \partial_r(q^\sigma f) = q^{\sigma-1/2} f_x. \tag{50}$$

For the last formula q, η are independent of r . For the first two, differentiation of x gives respectively $-\eta x q^{-D}/B$ and $x q^{-1}/(2B)$, proving every term.

In particular let the original radial moment bump be

$$b_e = q^{-(e+1)/2} \widehat{b}_e(x), \quad \int x^e \widehat{b}_e(x) dx = 1.$$

For any $k, a, b \geq 0$, its derivative is exactly

$$\partial_r^k \partial_z^a \partial_t^b b_e = q^{-(e+1+k)/2-aD-b} f_{e;k,a,b}(x, \eta). \quad (51)$$

Here the coefficient is given by a finite recursion: start at $\sigma = -(e+1+k)/2$ and $f = \widehat{b}_e^{(k)}$; repeat $f \leftarrow \mathcal{Z}_\sigma f$, $\sigma \leftarrow \sigma - D$, a times, and then $f \leftarrow \mathcal{T}_\sigma f$, $\sigma \leftarrow \sigma - 1$, b times. The result is $f_{e;k,a,b}$. This proves the identity by induction using (50). Smooth mixed derivatives commute, so this fixed order computes the stated derivative. Every coefficient is explicitly a finite rational expression in x, η , the retained constants, and derivatives of the specified bump, with nonzero denominator a power of B . On its compact x support and $|\eta| \leq 1$ its maximum is finite and computable directly from that expression. Thus (51) supplies the original power of q at every order. The pressure bump ρ uses exactly this recursion with $e = 0$. Likewise every derivative in (48) is explicit: $\partial_z^a \partial_t^b q = q^{1-aD-b} Q_{a,b}(\eta)$, where the same recursion starts with $\sigma = 1, f = 1$.

For completeness, the weighted cumulative bump has the exact form

$$\beta_e(r, z, t) = \int_0^r s^e b_e(s, z, t) ds = \int_0^{r/\sqrt{q}} x^e \widehat{b}_e(x) dx.$$

Its axial derivative is

$$\partial_z \beta_e = -\frac{qz}{2q} r^{e+1} b_e. \quad (52)$$

This follows by differentiating its upper endpoint, which is $-rqz/(2q^{3/2})$. Inserting the definition of b_e gives the displayed original physical factors. Therefore replacing the primitive of $F_e - b_e \partial_z J_e$ by the exact primitive of $F_e - \partial_z(b_e J_e)$ changes the stress coefficient by

$$-\frac{rqz}{2q} b_e J_e.$$

This is the moving-bump term required in the complete stress construction. Its sign and all further derivatives are fixed by (49)–(52); it is not set to zero by holding the physical source scale fixed during axial differentiation.

These calculations supply finite quantitative inputs for the actual source pressure and moment operations. They do not identify the distinct direct candidate with a completed infinite source correction sequence: the complete source-operation formulas retain the new velocity, covariance, pressure and nonlinear defects that such a step actually creates.

4 Physical derivative and full vector-force costs of the five-row mean operation

This section completes the finite mean operation on the actual current source state. All coordinates are the original right-handed cylindrical coordinates. The physical viscosity is the fixed number $\nu_{\text{NS}} > 0$. Write the full current velocity as $U = (b + \beta, V + v, G + \gamma) + w$, where w includes every current wave and curl correction. During the operation below this wave is held fixed. The three targets are the full current defects

$$\mathcal{P} = P_{\text{old}} + \Delta P, \quad \mathcal{J}_\theta = (J_\theta)_{\text{old}} + \Delta J_\theta, \quad \mathcal{J}_z = (J_z)_{\text{old}} + \Delta J_z. \quad (53)$$

Here the increments are the pressure-corrected actual wave increments of (47). If other finite source operations have already been performed, “old” means their actual entering state and the

increments are recomputed for the actual addition. Thus none of the following formulas substitutes an increment for a full target.

Retain, without a change of scale or profile constant,

$$\begin{aligned}
q - z^2 q^{2h} &= 1 - t, \quad z = q^D \eta, \quad x = r/\sqrt{q}, \quad A = \frac{1}{2} + h, \quad D = \frac{1}{2} - h, \\
0 < h < 1/100, \quad q > 0, \quad -1 < \eta < 1, \quad B(\eta) &= 1 - 2h\eta^2, \\
V &= q^{-A} a(\eta) x^{-1-2\lambda}, \quad G = 0 \quad \text{on the reserved mean patch}, \\
a(\eta) &= \frac{2^{1/2+\lambda} c_{\text{patch}}}{1 + \eta^2}, \quad \lambda > 0, \quad |a(\eta)| \geq a_0 > 0.
\end{aligned} \tag{54}$$

The symbol $a(\eta)$ in this display is a scalar base coefficient; below the bold symbol $\mathbf{a}$ denotes the vector increment. The original constant c_{patch} can have either nonzero sign.

4.1 Exact inverse profiles and the five physical rows

Use precisely the bumps and spacing of the five-row operation:

$$\varphi_j(x) = e^{-j d_b} \varphi_0(e^{-j d_b} x), \quad \mu_p = \int_0^\infty x^p \varphi_0(x) dx > 0, \quad t_p = e^{d_b p}, \quad d_b > 0.$$

The nonnegative nonzero smooth bump φ_0 and its three dilates have support in the reserved open interval $I_m \Subset (0, \infty)$. The substitution $x = e^{j d_b} \xi$ proves exactly $\int x^p \varphi_j(x) dx = \mu_p t_p^j$. Number matrix rows and columns starting at zero, and define

$$\begin{aligned}
V_\theta &= [t_p^j]_{p=2, -2-2\lambda, -2\lambda; j=0, 1, 2}, \quad B_\theta = V_\theta^{-1}, \\
V_z &= [t_p^j]_{p=1, 1-2\lambda; j=0, 1}, \quad B_z = V_z^{-1}, \\
A_\theta &= \text{diag}(\mu_2, 2a\mu_{-2-2\lambda}, -a\mu_{-2\lambda}) V_\theta, \quad A_z = \text{diag}(\mu_1, a\mu_{1-2\lambda}) V_z.
\end{aligned} \tag{55}$$

For three row values t_0, t_1, t_2 , subtraction of the first row from each later row, followed by factoring, gives $\det V_\theta = (t_1 - t_0)(t_2 - t_0)(t_2 - t_1)$. For two row values the determinant is $t_1 - t_0$. The original exponents in each displayed matrix are distinct because $\lambda > 0$, and the exponential with $d_b > 0$ is injective. The row factors are nonzero. This proves both inverses exist at the actual fixed source parameters. Explicitly each entry of either B is its corresponding transposed cofactor divided by the stated determinant; this is a finite formula in the original exponentials.

Define the following actual profiles, including their η dependence:

$$\begin{aligned}
V_P(x, \eta) &= -\frac{1}{2a(\eta)\mu_{-2-2\lambda}} \sum_{j=0}^2 B_\theta[j, 1] \varphi_j(x), \\
V_{J_z}(x, \eta) &= \frac{1}{a(\eta)\mu_{-2\lambda}} \sum_{j=0}^2 B_\theta[j, 2] \varphi_j(x), \\
G_{J_\theta}(x, \eta) &= -\frac{1}{a(\eta)\mu_{1-2\lambda}} \sum_{j=0}^1 B_z[j, 1] \varphi_j(x).
\end{aligned} \tag{56}$$

The signs, the factor two, and all three negative powers in the moment denominators are part of the formulas. Multiplying the columns by the exact rows in (55) proves the eight scalar profile

identities

$$\begin{array}{c|cc} & V_P & V_{J_z} \\ \hline \int x^2(\cdot) dx & 0 & 0 \\ 2a \int x^{-2-2\lambda}(\cdot) dx & -1 & 0 \\ -a \int x^{-2\lambda}(\cdot) dx & 0 & -1 \end{array} \quad \begin{array}{c|c} & G_{J_\theta} \\ \hline \int x(\cdot) dx & 0 \\ a \int x^{1-2\lambda}(\cdot) dx & -1 \end{array}. \quad (57)$$

There are eight entries in these two tables; each follows from $V_\theta B_\theta = I_3$ or $V_z B_z = I_2$, with the row factor left in place. The complete matrix identities include all nine and all four scalar inverse identities, respectively.

The source coefficients are

$$\mathbf{u} = A_\theta^{-1}(0, -q^{2A}\mathcal{P}, -q^{2A-1}\mathcal{J}_z)^T, \quad \mathbf{s} = A_z^{-1}(0, -q^{2A-3/2}\mathcal{J}_\theta)^T.$$

Since $(DV)^{-1} = V^{-1}D^{-1}$, substitution of these coefficients in the original physical increments gives

$$\delta v = q^A \mathcal{P} V_P + q^{A-1} \mathcal{J}_z V_{J_z}, \quad \delta \gamma = q^{A-3/2} \mathcal{J}_\theta G_{J_\theta}. \quad (58)$$

For example the fifth-row coefficient in $\mathbf{u}$ is $(-q^{2A-1}\mathcal{J}_z)/(-a\mu_{-2\lambda})$, which proves the positive sign in V_{J_z} . Multiplication by the original outer factor q^{-A} proves its exponent $A-1$. The other two terms follow by the same displayed diagonal products.

Changing variables $r = \sqrt{q}x$ in each integral, and using (57), now proves the full five equations

$$\begin{aligned} \int r^2 \delta v dr &= 0, & \int r \delta \gamma dr &= 0, & \int \frac{2V}{r} \delta v dr &= -\mathcal{P}, \\ \int r^2 (G \delta v + V \delta \gamma) dr &= -\mathcal{J}_\theta, \\ \int (2rG \delta \gamma - rV \delta v) dr &= -\mathcal{J}_z. \end{aligned} \quad (59)$$

Indeed the pressure term with V_P has scale $q^{-A}q^A = 1$; the angular flux with G_{J_θ} has scale $q^{3/2}q^{-A}q^{A-3/2} = 1$; and the fifth row with V_{J_z} has scale $qq^{-A}q^{A-1} = 1$. Every other term is one of the zero entries in the tables. The two homogeneous velocity moments use the first rows of those tables. This verifies the physical equations, including the single factor r multiplying $V\delta v$ in the fifth row.

4.2 Compact potential, induced radial velocity, and support

Define the combined potential profile

$$H_\theta(x, \eta) = \frac{1}{x} \int_0^x \xi G_{J_\theta}(\xi, \eta) d\xi. \quad (60)$$

Choose $0 < x_- < x_+$ such that the convex hull of all the bump supports lies in $[x_-, x_+] \subseteq I_m$. For $x < x_-$ the integral in (60) is zero. For $x > x_+$ it is the zero axial moment in (57). It follows that H_θ , together with every derivative, is supported in this fixed compact interval. Its smoothness at the two collars follows from the fact that the integrand is identically zero there and that its total integral is zero. In particular it extends smoothly by zero to $x \leq 0$.

The primitive of an individual φ_j is $\psi_j = x^{-1} \int_0^x \xi \varphi_j(\xi) d\xi$, which equals $\mu_1 e^{jd_b}/x$ past its support. Thus an individual primitive can have a tail. The exact identity $\sum_{j=0}^1 B_z[j, 1] e^{jd_b} = 0$ cancels these tails in $H_\theta = -(a\mu_{1-2\lambda})^{-1} \sum_j B_z[j, 1] \psi_j$. The identity holds for every η , so its

η derivatives also vanish. This proves compactness for the combined profile and all derivatives without assuming compactness for the summands.

The physical potential and radial increment are exactly

$$\begin{aligned}\Psi(r, z, t) &= \frac{1}{r} \int_0^r s \delta\gamma(s, z, t) ds = q^{A-1} \mathcal{J}_\theta H_\theta(x, \eta), \\ \delta\beta &= -\partial_z \Psi \\ &= -q^{A-1} (\partial_z \mathcal{J}_\theta) H_\theta - q^{A-1-D} \mathcal{J}_\theta \mathcal{Z}_{A-1} H_\theta.\end{aligned}\tag{61}$$

The substitution in the integral supplies one factor q and its outer $1/r$ supplies $q^{-1/2}$, giving the exponent $A - 1$. The operator in the second line is explicitly

$$\mathcal{Z}_\sigma f = \frac{2\sigma\eta f - \eta x f_x + (1 - \eta^2) f_\eta}{B(\eta)}.$$

The chain rule (50) proves the last line, including the η derivative of the original factor $1/a$. Direct differentiation of xH_θ gives $(\partial_x + x^{-1})H_\theta = G_{J_\theta}$, so $(\partial_r + r^{-1})\Psi = \delta\gamma$. Commuting the smooth physical r, z derivatives proves

$$(\partial_r + r^{-1})\delta\beta + \partial_z \delta\gamma = 0.\tag{62}$$

The three velocity increments and Ψ are auxiliary independent and angularly independent; their support lies in $\sqrt{q}[x_-, x_+]$, including any intervals between the bumps.

4.3 Every mixed physical derivative, with original scale powers

For clarity the original differential dictionary is

$$q_z = \frac{2\eta q^A}{B}, \quad q_t = -\frac{1}{B}, \quad \eta_z = \frac{q^{-D}(1 - \eta^2)}{B}, \quad \eta_t = \frac{D\eta q^{-1}}{B}, \quad \mathcal{T}_\sigma f = \frac{-\sigma f + \frac{1}{2}x f_x + D\eta f_\eta}{B}.$$

These equations follow by implicit differentiation of (54); in particular $1 - 2hz^2 q^{2h-1} = B$ and $D + 2h = A$. Differentiating $x = rq^{-1/2}$ supplies $x_z = -\eta x q^{-D}/B$ and $x_t = x q^{-1}/(2B)$. Consequently

$$\partial_r(q^\sigma F) = q^{\sigma-1/2} F_x, \quad \partial_z(q^\sigma F) = q^{\sigma-D} \mathcal{Z}_\sigma F, \quad \partial_t(q^\sigma F) = q^{\sigma-1} \mathcal{T}_\sigma F.$$

Here $F = F(x, \eta)$. These are physical derivatives of the moving scale; the time loss is the original one in this scale.

For $k, m, n \geq 0$ define $F_{\sigma;k,m,n}$ by the following finite recursion. Start with $f = \partial_x^k F$ and $s = \sigma - k/2$. Repeat $f \leftarrow \mathcal{Z}_s f$, $s \leftarrow s - D$ exactly m times; then repeat $f \leftarrow \mathcal{T}_s f$, $s \leftarrow s - 1$ exactly n times. The final f is $F_{\sigma;k,m,n}$. For any actual scalar target $X = X(z, t)$, the exact product formula is

$$\begin{aligned}\partial_r^k \partial_z^m \partial_t^n (q^\sigma X F) &= \sum_{i=0}^m \sum_{j=0}^n \binom{m}{i} \binom{n}{j} (\partial_z^i \partial_t^j X) \\ &\quad \cdot q^{\sigma-k/2-(m-i)D-(n-j)} F_{\sigma;k,m-i,n-j}(x, \eta).\end{aligned}\tag{63}$$

To prove it, first apply k radial derivatives, which do not act on X, q, η . The repeated product rule for the m axial and n time derivatives assigns i, j derivatives to X , with the two binomial multiplicities displayed. Apply the three chain rules above successively to the remaining factor. Each axial derivative lowers its current exponent by D , and each time derivative lowers it by one,

while its current exponent is the index in the next operator. This is precisely the stated recursion. Smooth mixed derivatives commute, so the chosen radial-then-axial- then-time order computes the derivative written in the formula.

This proves every mixed derivative of $\delta v, \delta \gamma, \Psi$ by using, respectively, the two terms and the one term in (58) and (61). It proves every derivative of $\delta \beta$ either by applying (63) to its two displayed terms or by

$$\partial_r^k \partial_z^m \partial_t^n \delta \beta = -\partial_r^k \partial_z^{m+1} \partial_t^n (q^{A-1} \mathcal{J}_\theta H_\theta). \quad (64)$$

This equality retains, in particular, the derivative of the full target and the additional original factor q^{-D} .

The recursion is an explicit finite rational expression in x, η, h , the retained profile derivatives, and powers of B^{-1} . The other profile constants are exactly the inverse Vandermonde entries and moments in (56). In fact the reciprocal base coefficient has the exact derivatives

$$\partial_\eta^s (a^{-1}) = \frac{1}{2^{1/2+\lambda} c_{\text{patch}}} \begin{cases} 1 + \eta^2 & s = 0, \\ 2\eta & s = 1, \\ 2 & s = 2, \\ 0 & s \geq 3. \end{cases} \quad (65)$$

Also $\varphi_j^{(k)}(x) = e^{-(k+1)jdb} \varphi_0^{(k)}(e^{-jdb}x)$. These two formulas give every mixed derivative of the three profiles without an unspecified inverse-matrix derivative constant.

For a similarly explicit derivative of H_θ , write $F_0(x, \eta) = \int_0^x \xi G_{J_\theta}(\xi, \eta) d\xi$. For $\ell \geq 1$ the fundamental theorem of calculus and the product rule give

$$\partial_x^\ell F_0 = x \partial_x^{\ell-1} G_{J_\theta} + (\ell - 1) \partial_x^{\ell-2} G_{J_\theta},$$

where the second term is defined to be zero for $\ell = 1$. Therefore, for $x > 0$,

$$\partial_x^k \partial_\eta^s H_\theta = \sum_{\ell=0}^k \binom{k}{\ell} (-1)^{k-\ell} (k-\ell)! x^{-1-k+\ell} \partial_x^\ell \partial_\eta^s F_0. \quad (66)$$

The $\ell = 0$ factor is the explicitly specified integral; every other factor is the displayed profile derivative. For example its absolute integral is at most $\int_{x_-}^{x_+} \xi |\partial_\eta^s G_{J_\theta}(\xi, \eta)| d\xi$. This supplies finite computable profile costs on the exact support. The bound $B \geq 1 - 2h > 0$, together with $x \geq x_- > 0$, proves that each fixed recursively specified coefficient is bounded on $[x_-, x_+] \times [-1, 1]$. The original moments, spacing, h , c_{patch} , and λ remain in those coefficients.

4.4 Explicit costs in full target derivative arrays

For $I = (m, n)$, put $I_z = (m + 1, n)$. Use the original actual-shape arrays $\mathcal{B}_{e,I}$ from (45), with their two retained coupled amplitude components and old-wave cross terms. Define the full target majorants

$$\begin{aligned} P_I &= |\partial_z^m \partial_t^n P_{\text{old}}| + \mathcal{B}_{-1,I} + \frac{1}{2} \mathcal{B}_{0,I_z}, \\ J_{\theta,I} &= |\partial_z^m \partial_t^n (J_\theta)_{\text{old}}| + \frac{1}{2} \mathcal{B}_{2,I}, \\ J_{z,I} &= |\partial_z^m \partial_t^n (J_z)_{\text{old}}| + \mathcal{B}_{1,I} + \frac{1}{4} \mathcal{B}_{2,I_z}. \end{aligned} \quad (67)$$

The triangle inequality in (53) and the proved increment bounds (48) give $|\partial^I \mathcal{P}| \leq P_I$, $|\partial^I \mathcal{J}_\theta| \leq J_{\theta,I}$, and $|\partial^I \mathcal{J}_z| \leq J_{z,I}$. Thus the old absolute derivatives appear explicitly in every cost.

For a nonnegative target array $\mathbf{X}$, define the following pointwise, fully specified finite sum:

$$\mathfrak{C}_{\sigma,F;k,m,n}[\mathbf{X}](r, z, t) = \sum_{i=0}^m \sum_{j=0}^n \binom{m}{i} \binom{n}{j} \mathbf{X}_{(i,j)}(z, t) \cdot q^{\sigma-k/2-(m-i)D-(n-j)} |F_{\sigma;k,m-i,n-j}(x, \eta)|. \quad (68)$$

All powers are powers of the actual positive q , even when their exponents are negative. Define

$$\begin{aligned} \mathbf{A}_{\theta;k,m,n} &= \mathfrak{C}_{A,V_P;k,m,n}[\mathbf{P}] + \mathfrak{C}_{A-1,V_{J_z};k,m,n}[\mathbf{J}_z], \\ \mathbf{A}_{z;k,m,n} &= \mathfrak{C}_{A-3/2,G_{J_\theta};k,m,n}[\mathbf{J}_\theta], \\ \mathbf{S}_{k,m,n} &= \mathfrak{C}_{A-1,H_\theta;k,m,n}[\mathbf{J}_\theta], \\ \mathbf{A}_{r;k,m,n} &= \mathfrak{C}_{A-1,H_\theta;k,m,n}[\mathbf{J}_\theta^{+z}] + \mathfrak{C}_{A-1-D,\mathcal{Z}_{A-1}H_\theta;k,m,n}[\mathbf{J}_\theta], \quad (\mathbf{J}_\theta^{+z})_{(i,j)} = \mathbf{J}_{\theta,(i+1,j)}. \end{aligned} \quad (69)$$

Equations (63)–(61) and the triangle inequality prove, for every k, m, n ,

$$\begin{aligned} |\partial_r^k \partial_z^m \partial_t^n \delta v| &\leq \mathbf{A}_{\theta;k,m,n}, & |\partial_r^k \partial_z^m \partial_t^n \delta \gamma| &\leq \mathbf{A}_{z;k,m,n}, \\ |\partial_r^k \partial_z^m \partial_t^n \Psi| &\leq \mathbf{S}_{k,m,n}, & |\partial_r^k \partial_z^m \partial_t^n \delta \beta| &\leq \mathbf{A}_{r;k,m,n}. \end{aligned} \quad (70)$$

The alternative upper bound $\mathbf{S}_{k,m+1,n}$ for the last quantity follows directly from (64). For example the zero-order costs read

$$\begin{aligned} |\delta v| &\leq q^A \mathbf{P}_{0,0} |V_P| + q^{A-1} \mathbf{J}_{z,0,0} |V_{J_z}|, \\ |\delta \gamma| &\leq q^{A-3/2} \mathbf{J}_{\theta,0,0} |G_{J_\theta}|, & |\Psi| &\leq q^{A-1} \mathbf{J}_{\theta,0,0} |H_\theta|, \\ |\delta \beta| &\leq q^{A-1} \mathbf{J}_{\theta,1,0} |H_\theta| + q^{A-1-D} \mathbf{J}_{\theta,0,0} |\mathcal{Z}_{A-1} H_\theta|. \end{aligned}$$

These display separately the target derivative and moving-scale radial cost. Replacing each profile value in (68) by its maximum over its fixed x, η support supplies a pointwise bound depending only on z, t and the stated fixed profile data.

The same formula gives weighted radial costs without a loss hidden in a constant. For any real e , replace the final profile absolute value in (68) by

$$q^{(e+1)/4} \left(\int_{x_-}^{x_+} x^e |F_{\sigma;k,m-i,n-j}(x, \eta)|^2 dx \right)^{1/2}.$$

Call the resulting sum $\mathfrak{C}^{[e]}$. The exact measure identity $r^e dr = q^{(e+1)/2} x^e dx$, followed by the triangle inequality in this weighted L^2 space, proves all four bounds (70) with the left sides replaced by their $[e]$ norms and the right sides by the corresponding $\mathfrak{C}^{[e]}$ sums. The support is bounded away from zero, so these integrals are finite for every fixed real e .

4.5 Actual pressure reconstruction and the surviving defects

Set $\mathbf{a} = (a_r, a_\theta, a_z) = (\delta \beta, \delta v, \delta \gamma)$. Before pressure is changed, let δg_r denote the exact change in the source's angularly averaged negative radial residual. The angular average here retains auxiliary dependence. The actual pressure-defect change is

$$\delta P_{\text{step}} = \int_0^\infty \overline{\delta g_r} dr = P_{\text{new}} - \mathcal{P}. \quad (71)$$

This is the defect change caused by the present mean operation; it is different from ΔP in (53). The source pressure bump is still $\rho = q^{-1/2}\hat{\rho}(x)$, $\int \hat{\rho} dx = 1$. The actual pressure increment for this operation is

$$\delta p = \mathcal{I}_c f, \quad f = \delta g_r - \rho \delta P_{\text{step}}. \quad (72)$$

The original physical shifted primitive is

$$\begin{aligned} \mathcal{I}f(r, Y) &= \int_0^r f(s, Y + (s^{d_r} - r^{d_r})v_r, z, t) ds, \\ \mathcal{J}f(r, Y) &= \int_0^\infty f(s, Y + (s^{d_r} - r^{d_r})v_r, z, t) ds, \\ \mathcal{I}_c f &= \mathcal{I}f - \chi_m \mathcal{J}f, \\ d_r &= 2((1+h)\rho_g - h\kappa_s), \quad \rho_g = \frac{\log(4 - \sqrt{2})}{\log(4 + \sqrt{2})}, \quad \kappa_s = 10^{-5}, \quad v_r = (1, -(\sqrt{2} - 1)). \end{aligned}$$

Here χ_m is the actual inner-zero, outer-one source cutoff; the two suppressed arguments of each integrand are exactly z, t . The pressure entering the physical force is the physical evaluation of this source function, with its original phase evaluation and all its derivatives. Thus it includes any nonzero auxiliary modes of the source reconstruction.

To verify the reconstruction identity, the source physical radial operator $\mathcal{D}_r$ cancels the derivative of the term $-r^{d_r}v_r$ in the shift. Endpoint differentiation gives $\mathcal{D}_r \mathcal{I}f = f$ and $\mathcal{D}_r \mathcal{J}f = 0$, and hence

$$\mathcal{D}_r \delta p - \delta g_r = -\rho \delta P_{\text{step}} - (\partial_r \chi_m) \mathcal{J}f. \quad (73)$$

Haar translation invariance and (71) give $\overline{\mathcal{J}f} = 0$, but this does not set $\mathcal{J}f$ itself to zero. In the inner collar $\mathcal{I}f = 0$ and $\chi_m = 0$; in the outer collar $\mathcal{I}f = \mathcal{J}f$ and $\chi_m = 1$. This proves compact support of the actual pressure and all derivatives. In particular its gradient is retained in the force calculation below.

For completeness the exact remaining source terms can be written entirely in the original current mean fields. With $\Delta_0 = \mathcal{D}_r^2 + r^{-1}\mathcal{D}_r + \partial_z^2$, direct expansion of the radial conservative source gives

$$\begin{aligned} R_g &:= \delta g_r - \frac{2V}{r} \delta v \\ &= -\partial_t \delta \beta - (\mathcal{D}_r + r^{-1})(2b\delta\beta + 2\beta\delta\beta + (\delta\beta)^2) \\ &\quad - \partial_z(b\delta\gamma + G\delta\beta + \beta\delta\gamma + \gamma\delta\beta + \delta\beta\delta\gamma) \\ &\quad + r^{-1}(2v\delta v + (\delta v)^2) + \nu_{\text{NS}}(\Delta_0 - r^{-2})\delta\beta. \end{aligned} \quad (74)$$

Indeed the radial conservative quadratic expression before the change is $-(\mathcal{D}_r + r^{-1})(b + \beta)^2 - \partial_z((b + \beta)(G + \gamma)) + (V + v)^2/r$, together with the wave covariance, time term, and viscosity. Subtract it from its value after the mean change. The unchanged wave covariance cancels; each remaining quadratic product expands to the terms displayed. The term $2V\delta v/r$ has been written separately in the definition of R_g . The ordinary time derivative on the increment is exact because it is auxiliary independent.

The five physical rows then give the exact new defects

$$\begin{aligned} P_{\text{new}} &= \int \overline{R_g} dr, \\ (J_\theta)_{\text{new}} &= \int r^2 \overline{\gamma\delta v + v\delta\gamma + \delta\gamma\delta v} dr, \\ (J_z)_{\text{new}} &= \int r^2 \overline{2\gamma\delta\gamma + (\delta\gamma)^2} dr - \frac{1}{2} \int r^2 \overline{R_g} dr. \end{aligned} \quad (75)$$

For the first line add $\mathcal{P}$ to $\int (2V\delta v/r + \overline{R_g}) dr$ and use the third row. For the second, expanding $(G+\gamma+\delta\gamma)(V+v+\delta v) - (G+\gamma)(V+v)$ gives base-linear terms $G\delta v + V\delta\gamma$, cancelled by the fourth row, and exactly the three remaining terms displayed. For the third, expand $(G+\gamma+\delta\gamma)^2 - (G+\gamma)^2$ and the radial-source moment $-\frac{1}{2} \int r^2 \delta g_r dr$; the base-linear terms are the fifth row. This proves all three equalities. Neither their right sides nor the cutoff term in (73) has been removed.

4.6 Exact full cylindrical physical force increment

For the actual physical fields, put $\mathcal{R}(U, p) = \partial_t U + (U \cdot \nabla)U - \nu_{\text{NS}} \Delta U + \nabla p$. Ordinary derivatives in this formula are derivatives after the physical source evaluation; they therefore include all phase, frame, cutoff and scale derivatives of the actual full U . Expanding the two quadratic products proves the vector identity

$$\delta \mathcal{R} = \partial_t \mathbf{a} + (U \cdot \nabla) \mathbf{a} + (\mathbf{a} \cdot \nabla) U + (\mathbf{a} \cdot \nabla) \mathbf{a} - \nu_{\text{NS}} \Delta \mathbf{a} + \nabla \delta p. \quad (76)$$

The pressure here is precisely (72).

To derive every cylindrical component, the basis derivatives are $\partial_\theta e_r = e_\theta$, $\partial_\theta e_\theta = -e_r$, and $\partial_\theta e_z = 0$. Thus for two arbitrary vector fields X, Y the component vector of $(X \cdot \nabla)Y$ is

$$X_r \partial_r Y + \frac{X_\theta}{r} \partial_\theta Y + X_z \partial_z Y + \frac{X_\theta}{r} (-Y_\theta, Y_r, 0).$$

Apply this equality to $(U, \mathbf{a})$, $(\mathbf{a}, U)$, and $(\mathbf{a}, \mathbf{a})$. The scalar components of $\mathbf{a}$ have zero angular derivative, whereas those of the actual U generally do not. Writing $L_0 = \partial_r^2 + r^{-1} \partial_r + \partial_z^2$, the complete result is

$$\begin{aligned} \delta \mathcal{R}_r = & \partial_t a_r + U_r \partial_r a_r + U_z \partial_z a_r + a_r \partial_r U_r + a_z \partial_z U_r + \frac{a_\theta}{r} \partial_\theta U_r \\ & + a_r \partial_r a_r + a_z \partial_z a_r - \frac{2U_\theta a_\theta + a_\theta^2}{r} - \nu_{\text{NS}} (L_0 - r^{-2}) a_r + \partial_r \delta p, \end{aligned} \quad (77)$$

$$\begin{aligned} \delta \mathcal{R}_\theta = & \partial_t a_\theta + U_r \partial_r a_\theta + U_z \partial_z a_\theta + a_r \partial_r U_\theta + a_z \partial_z U_\theta + \frac{a_\theta}{r} \partial_\theta U_\theta \\ & + a_r \partial_r a_\theta + a_z \partial_z a_\theta + \frac{U_r a_\theta + a_r U_\theta + a_r a_\theta}{r} - \nu_{\text{NS}} (L_0 - r^{-2}) a_\theta + \frac{1}{r} \partial_\theta \delta p, \end{aligned} \quad (78)$$

$$\begin{aligned} \delta \mathcal{R}_z = & \partial_t a_z + U_r \partial_r a_z + U_z \partial_z a_z + a_r \partial_r U_z + a_z \partial_z U_z + \frac{a_\theta}{r} \partial_\theta U_z \\ & + a_r \partial_r a_z + a_z \partial_z a_z - \nu_{\text{NS}} L_0 a_z + \partial_z \delta p. \end{aligned} \quad (79)$$

The self terms in the radial equation include $-a_\theta^2/r$; those in the angular equation include $a_r a_\theta/r$. The old-wave cross terms occur in the actual U and in its angular derivatives in all three equations. The mean pressure is angularly invariant, so its angular derivative is zero for this source operation; its radial and axial derivatives are the actual reconstructed ones and are generally nonzero.

For the viscosity formula, applying the scalar cylindrical Laplacian to the moving basis yields, for a general vector field X ,

$$\Delta X = (\Delta_s X_r - r^{-2} X_r - 2r^{-2} \partial_\theta X_\theta, \Delta_s X_\theta - r^{-2} X_\theta + 2r^{-2} \partial_\theta X_r, \Delta_s X_z), \quad \Delta_s = L_0 + r^{-2} \partial_\theta^2.$$

This follows by differentiating the two basis identities twice and applying the product rule. Substituting the angularly independent components of $\mathbf{a}$ gives exactly $((L_0 - r^{-2})a_r, (L_0 - r^{-2})a_\theta, L_0 a_z)$, proving all three displayed viscous terms. The radial expression before the pressure term, when angularly averaged in the extended source domain and negated, is precisely δg_r : this follows from the definition of the negative radial source residual and linearity of the average. It is also an explicit formula for the input to (72) in terms of the same full velocity and mean increments.

4.7 Finite all-order force costs with every inverse-radius term

Here are explicit derivative bounds for all three components, including angular derivatives of the actual current velocity. Let $\alpha = (k, \ell, m, n) \in \mathbb{N}^4$ index the scalar component derivative $\partial^\alpha = \partial_r^k \partial_\theta^\ell \partial_z^m \partial_t^n$, and let e_r, e_θ, e_z, e_t be the four coordinate multi-indices. Extend the arrays in (69) by

$$\mathbf{A}_{i,\alpha} = \begin{cases} \mathbf{A}_{i;k,m,n} & \ell = 0, \\ 0 & \ell > 0, \end{cases} \quad \mathbf{U}_{i,\alpha} = |\partial^\alpha U_i|.$$

The pressure array Π will be the following explicit majorant for the derivatives of the actual specified function (72), after physical evaluation.

The original physical radial phase satisfies $Y_{\text{phys}}(s, z, t) - Y_{\text{phys}}(r, z, t) = (s^{d_r} - r^{d_r})v_r$. Substituting this exact equality in the two shifted integrals proves the evaluated formula

$$\begin{aligned} \delta p(r, z, t) &= \int_0^r f_{\text{phys}}(s, z, t) ds - \chi_m(r, z, t) \int_0^\infty f_{\text{phys}}(s, z, t) ds, \\ f_{\text{phys}} &= (\delta g_r)_{\text{phys}} - \rho \delta P_{\text{step}}. \end{aligned} \tag{80}$$

Every original axial and temporal phase dependence is retained in f_{phys} . The second integral is independent of r after this evaluation. This equality does not assert that it is zero.

Fix any compact interior physical domain for the finite operation, and choose one physical interval $[R_-, R_+] \subseteq (0, \infty)$ containing the supports of δg_r and ρ for that domain and every auxiliary phase. Put $L_R = R_+ - R_-$, with these actual endpoints kept fixed during differentiation. The smooth zero extensions imply that all their derivatives have support in the same interval. Define $\mathbf{G}_{k,m,n}(z, t)$ as the supremum over $r \in [R_-, R_+]$ and the original auxiliary phases of the angular average of the following nonnegative expression: the radial cost $\mathbf{F}_{r;(k,0,m,n)}$ in (86) with its last pressure term deleted. This definition involves only $\mathbf{A}$, the actual current velocity derivatives, ν_{NS} , and the inverse-radius arrays; it has no dependence on Π . When auxiliary phases are retained, the velocity derivative arrays mean the corresponding full lifted physical derivatives before evaluation. The chain rule for source evaluation identifies these with derivatives of the evaluated fields at each phase. The force identities and the derivative proof below therefore imply

$$|\partial_r^k \partial_z^m \partial_t^n (\delta g_r)_{\text{phys}}| \leq \mathbf{G}_{k,m,n}, \quad |\partial_z^m \partial_t^n \delta P_{\text{step}}| \leq \mathbf{D}_{m,n} := L_R \mathbf{G}_{0,m,n}.$$

For the first inequality use the negative angularly averaged radial force before pressure, as proved above, and the triangle inequality under that average. For the second, differentiate (71) under its compact radial integral, commute axial and temporal differentiation with Haar average, and integrate the same bound over the interval of width L_R . The Haar integral of each original constant torus direction derivative is zero by periodicity, which justifies that commutation for the full physical time derivative as well.

The derivative of the pressure bump is already an explicit original- scale coefficient in (51). Define its exact support supremum

$$\mathbf{H}_{k,m,n}(z, t) = q^{-(1+k)/2-mD-n} \sup_{r \in [R_-, R_+]} |f_{0;k,m,n}(r/\sqrt{q}, \eta)|,$$

where $f_{0;k,m,n}$ is the recursion starting with the actual $\hat{\rho}$, rather than another normalized bump. It follows that $|\partial_r^k \partial_z^m \partial_t^n \rho| \leq \mathbf{H}_{k,m,n}$. Set

$$\mathbf{E}_{k,m,n} = \mathbf{G}_{k,m,n} + \sum_{i=0}^m \sum_{j=0}^n \binom{m}{i} \binom{n}{j} \mathbf{H}_{k,m-i,n-j} \mathbf{D}_{i,j}. \tag{81}$$

Since δP_{step} has no radial dependence, all k radial derivatives in its product with ρ act on ρ . The repeated axial and temporal product rule proves $|\partial_r^k \partial_z^m \partial_t^n f_{\text{phys}}| \leq E_{k,m,n}$ on the full interval.

Write $X_{k,i,j}(r, z, t) = |\partial_r^k \partial_z^i \partial_t^j \chi_m(r, z, t)|$, retaining the actual cutoff and its moving-scale derivatives. Define

$$\begin{aligned} \Pi_{(k,0,m,n)} &= \begin{cases} L_R E_{0,m,n} & k = 0, \\ E_{k-1,m,n} & k \geq 1 \end{cases} \\ &+ L_R \sum_{i=0}^m \sum_{j=0}^n \binom{m}{i} \binom{n}{j} X_{k,i,j} E_{0,m-i,n-j}, \quad \Pi_{(k,\ell,m,n)} = 0 \quad (\ell > 0). \end{aligned} \quad (82)$$

This proves the explicit bound $|\partial^\alpha \delta p| \leq \Pi_\alpha$. In fact differentiating (80) yields the exact first-integral derivative

$$\partial_r^k \partial_z^m \partial_t^n \int_0^r f_{\text{phys}}(s, z, t) ds = \begin{cases} \int_0^r \partial_z^m \partial_t^n f_{\text{phys}}(s, z, t) ds & k = 0, \\ \partial_r^{k-1} \partial_z^m \partial_t^n f_{\text{phys}}(r, z, t) & k \geq 1. \end{cases}$$

The absolute values are bounded by the first term in (82). In the cutoff product all radial derivatives act on χ_m , whereas the axial and temporal derivatives distribute with the two displayed binomial factors. Every derivative of its radial integral is bounded by $L_R E_{0,m-i,n-j}$. This proves its second term, with every cutoff derivative and the actual support width retained. Angular invariance proves the stated zero entries. This construction first computes the pressure-free G , then D, H, E, Π , and finally the full force array below, so the pressure budget is a finite calculation without circularly assuming a pressure bound.

For nonnegative arrays define their exact Leibniz convolution and derivative shift by

$$(X \star Y)_\alpha = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} X_\beta Y_{\alpha-\beta}, \quad (S_\delta X)_\alpha = X_{\alpha+\delta}. \quad (83)$$

Products with three factors use either parenthesization; multiplying the binomial coefficients gives $\alpha! / (\beta! \gamma! (\alpha - \beta - \gamma)!)$, proving the two parenthesizations agree. For the exact inverse-radius arrays put

$$(R_j)_\alpha = \begin{cases} (j)_k r^{-j-k} & \ell = m = n = 0, \\ 0 & \text{otherwise,} \end{cases} \quad (j)_0 = 1, \quad (j)_k = j(j+1) \cdots (j+k-1), \quad j = 1, 2. \quad (84)$$

Induction gives $\partial_r^k r^{-j} = (-1)^k (j)_k r^{-j-k}$, which proves that R_j is its exact absolute derivative array. No radial derivative of an inverse-radius coefficient is omitted in the following costs.

For $i = r, \theta, z$ define the common transport cost

$$\begin{aligned} T_i &= S_{e_t} A_i + U_r \star S_{e_r} A_i + U_z \star S_{e_z} A_i + A_r \star S_{e_r} U_i + A_z \star S_{e_z} U_i \\ &+ R_1 \star A_\theta \star S_{e_\theta} U_i + A_r \star S_{e_r} A_i + A_z \star S_{e_z} A_i, \end{aligned} \quad (85)$$

and the exact finite upper-cost arrays

$$\begin{aligned} F_r &= T_r + 2R_1 \star U_\theta \star A_\theta + R_1 \star A_\theta \star A_\theta \\ &+ \nu_{\text{NS}} (S_{2e_r} A_r + R_1 \star S_{e_r} A_r + S_{2e_z} A_r + R_2 \star A_r) + S_{e_r} \Pi, \\ F_\theta &= T_\theta + R_1 \star (U_r \star A_\theta + A_r \star U_\theta + A_r \star A_\theta) \\ &+ \nu_{\text{NS}} (S_{2e_r} A_\theta + R_1 \star S_{e_r} A_\theta + S_{2e_z} A_\theta + R_2 \star A_\theta) + R_1 \star S_{e_\theta} \Pi, \\ F_z &= T_z + \nu_{\text{NS}} (S_{2e_r} A_z + R_1 \star S_{e_r} A_z + S_{2e_z} A_z) + S_{e_z} \Pi. \end{aligned} \quad (86)$$

Every addition and scalar multiplication here is entrywise. For every finite α , these arrays prove the full physical component inequalities

$$|\partial^\alpha \delta \mathcal{R}_i| \leq F_{i,\alpha}, \quad i = r, \theta, z. \quad (87)$$

Indeed the repeated Leibniz rule gives $|\partial^\alpha(fg)| \leq (X \star Y)_\alpha$ whenever the respective derivative absolute values are bounded by X, Y . The three-factor version is its iterated rule, proved by the explicit multinomial coefficient above. Apply these rules term by term to (77)–(79), using (70) for the increment, the actual derivatives for $U, \delta p$, and (84) for each inverse radius. The constant viscosity passes through every derivative. The three resulting sums are precisely (85)–(86), proving the claim at every order.

These are component derivatives in the original orthonormal basis. For an explicit bound on the actual Cartesian vector after angular differentiation, the moving basis gives

$$\left| \partial_r^k \partial_\theta^\ell \partial_z^m \partial_t^n \delta \mathcal{R} \right| \leq \sum_{j=0}^{\ell} \binom{\ell}{j} (F_{r;(k,j,m,n)} + F_{\theta;(k,j,m,n)}) + F_{z;(k,\ell,m,n)}.$$

To prove it, expand each angular derivative by the binomial rule between a scalar component and its basis vector. Every derivative of e_r, e_θ has norm one, whereas the derivatives of e_z of positive order vanish. The triangle inequality gives exactly the displayed bound. Thus the component formulas also provide a bound for the physical vector, with its angular basis differentiation.

There is also an exact recursion for every Cartesian derivative. Let O_θ be the orthogonal matrix with columns (e_r, e_θ, e_z) , and let $J(f_r, f_\theta, f_z) = (-f_\theta, f_r, 0)$. Use $x_C = r \cos \theta, y_C = r \sin \theta$ for the physical Cartesian coordinates, distinct from the original scaled radius $x = r/\sqrt{q}$. Define the operators on component vectors

$$\begin{aligned} \mathcal{X}f &= \cos \theta \partial_r f - \frac{\sin \theta}{r} (\partial_\theta f + Jf), \\ \mathcal{Y}f &= \sin \theta \partial_r f + \frac{\cos \theta}{r} (\partial_\theta f + Jf), \quad \mathcal{Z}f = \partial_z f. \end{aligned} \quad (88)$$

The scalar coordinate rules $\partial_{x_C} = \cos \theta \partial_r - r^{-1} \sin \theta \partial_\theta$ and $\partial_{y_C} = \sin \theta \partial_r + r^{-1} \cos \theta \partial_\theta$, together with $\partial_\theta O_\theta = O_\theta J$, give exactly $\partial_{x_C}(O_\theta f) = O_\theta \mathcal{X}f$, $\partial_{y_C}(O_\theta f) = O_\theta \mathcal{Y}f$, and $\partial_z(O_\theta f) = O_\theta \mathcal{Z}f$. Applying these identities repeatedly proves

$$\partial_{x_C}^i \partial_{y_C}^j \partial_z^k \partial_t^n (O_\theta f) = O_\theta \mathcal{X}^i \mathcal{Y}^j \mathcal{Z}^k \partial_t^n f.$$

The order written is a fixed composition order and is exact because it was obtained from commuting smooth Cartesian derivatives. For a fully explicit majorant recursion, if C_r, C_θ, C_z majorize the component derivative arrays of f , set $(JC)_r = C_\theta$, $(JC)_\theta = C_r$, $(JC)_z = 0$. Let $c_\alpha = |\partial^\alpha \cos \theta|$, $s_\alpha = |\partial^\alpha \sin \theta|$, whose entries vanish unless $k = m = n = 0$ and whose remaining entries are the actual derivatives of these elementary functions. Then one $\mathcal{X}$ application is bounded componentwise by

$$c \star S_{e_r} C_i + s \star R_1 \star (S_{e_\theta} C_i + (JC)_i),$$

and one $\mathcal{Y}$ application by the same expression with c, s interchanged. A $\mathcal{Z}$ or time application is the array shift S_{e_z} or S_{e_t} . These rules follow from the exact product rule (83). Starting with F , iterating them the indicated finite number of times, and taking the sum of the three resulting zero-order component bounds gives a finite bound for each stated Cartesian vector derivative. Thus all Cartesian inverse-radius and angular-basis costs are also explicitly supplied.

All costs above are finite for the actual finite state on each compact interior physical domain. Their dependence on the original scale, target derivatives, inverse-radius factors, viscosity, full current velocity, and reconstructed pressure is explicit. The exact nonlinear defects (75) remain after this operation. No uniform estimate over an increasing family of source stages, no singular-time limit, and no infinite correction gain is asserted by these finite derivative identities and costs.